# Understanding What Drives Student–Teacher Ratios Across Tamil Nadu

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Introduction: In India, Tamil Nadu is one of the most socioeconomically diverse states and it is also characterized by great variation in rural development indicators among the districts and villages. Essential to development planning is the analysis of how health, education, and infrastructure factors impact upon the results of education systems. This research project focuses on analyzing student-teacher ratios at the village level. To this end, we will use the Mission Antyodaya database, which is an example of a national level database that has been developed to collect information about the assets, public services, and socioeconomic status of the villages throughout the country. The student-to-teacher ratio (STR) has been determined to represent a primary measure of educational service delivery. In essence, a lower STR would indicate a higher level of availability of teachers and a stronger commitment to supporting students' learning. The districts of this state vary greatly in population density, amenities, health care services, economic activity, and geographic features. Village-level statistical modeling could therefore show whether there are systematic influences of certain themes, like health facilities, educational infrastructure, and village amenities, on STR across districts. Two forms of regression modeling are utilized in this project, namely district level and state level. District level provides information about the variation within each state, while the state level model is used with a district dummy variable which provides insight on how district's structural characteristics differ from one another, and if any district performs better or worse than predicted by taking into consideration the observable variables. The information obtained through these models may provide an avenue to identify what type of investment is needed at the local villages to enhance their educational environment through better health access, improved educational infrastructure, or the implementation of adequate basic amenities. In addition, by comparing the results of the stepwise regression to the results obtained from the full model, we will gain additional insights about how different modeling techniques have affected variable selection and the overall interpretation of the results.

METHODOLOGY

The analysis are done based on the given guidelines for the project for the mission antyodaya dataset of Tamil Nadu. First we began with the merging SHRIDs with the district name, data cleaning, and then removing NA or any unrealistic values. This allowed us to get more accurate data estimates. Then the STRs were calculated for villages where both student and teacher numbers were accurately reported after the data was filtered Thereafter, within each districts, villages were randomly divided into a training 70% of the villages and a test data set 30% of the villages to complete the training and testing aspects of calculating district-level predictions. The training data set were used for the fit models and test data set were used for predictive accuracy.

For all districts for which STR values were calculated, run the thematic regressions using health, education, facilities, and other relevant variables to identify which variables were the main predictors that influence STR, and to record the $R^2$ and RMSE values. This was used to create a combined regression model, using stepwise regration to identify the best predictor variables without increasing the model's ability to predict beyond what is actually known from the data. Then from this final regression model using residual-fit plots for both the training and test datasets, we evaluated the accuracy of predictions

We used district dummy variables to control for variations in the structure of the villages included in the model, analyzed the final pooled regression model for villages within each district to capture structural differences between villages within one district and another, and used bootstrapping to calculate 95% confidence intervals around the coefficients of the most important variables for the predicted outcomes.

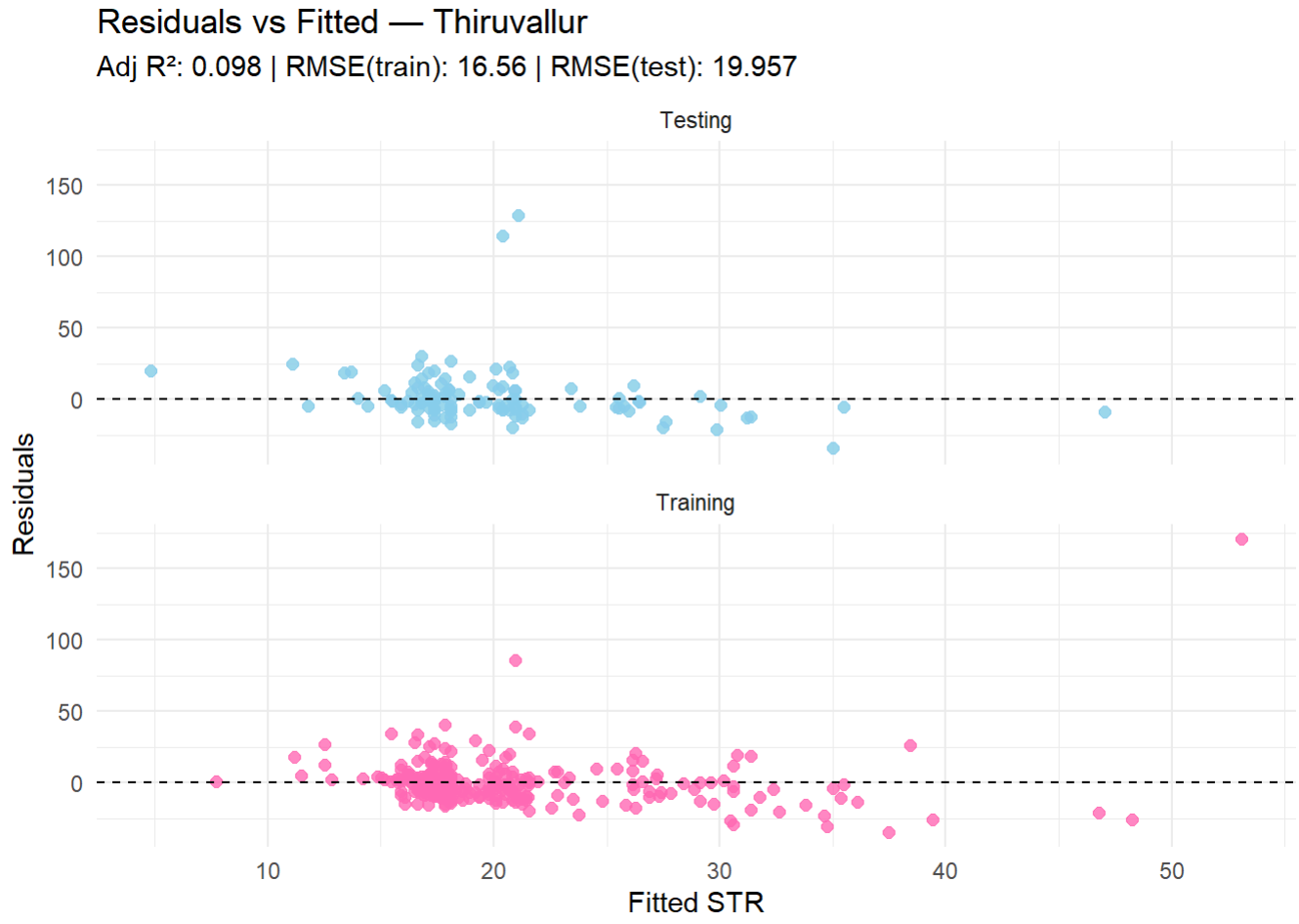
#Part A — District-Wise Village-Level Regression Analysis

Significant Variables

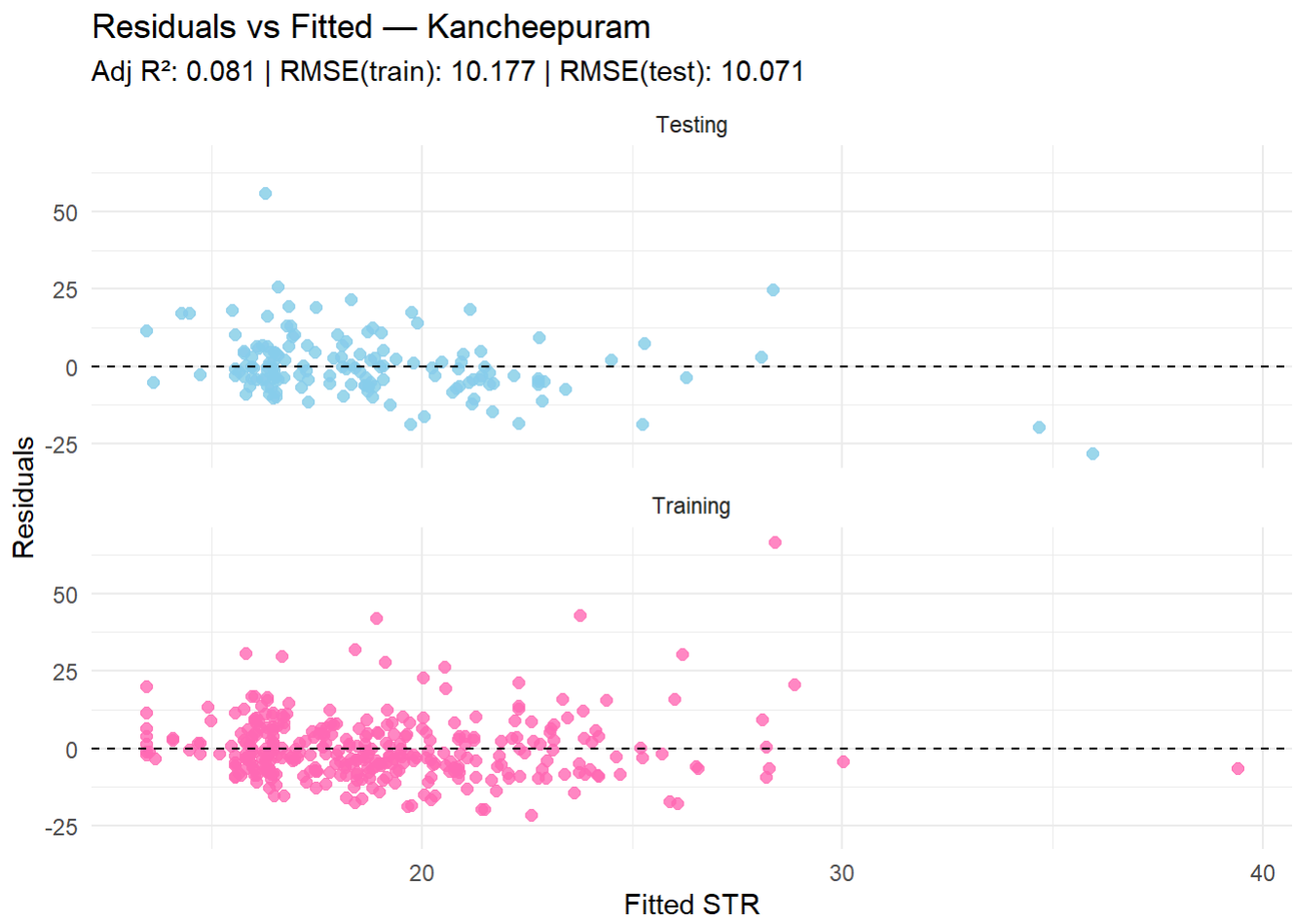
| Variable |
|---|
| availability_of_high_school |
| availability_of_middle_school |
| availability_of_mid_day_meal_sch |
| any_primary_sch_toilet |
| is_aanganwadi_centre_available |
| availability_of_ssc_school |
| total_hhd_availing_pension_under |
| sub_centre |
| total_no_of_pregnant_women |
| total_no_of_pregnant_women_recei |
| phc |
| total_childs_aged_0_to_3_years_r |
| closed_drainage |
| is_post_office_available |
| total_hhd_having_piped_water_con |
| regular_market |
| total_trained_trainee_under_skil |
| no_of_children_not_attending_sch |
| total_no_of_elected_representati |
| total_hhd_with_clean_energy |
| availability_of_railway_station |
| total_childs_categorized_non_stu |

We started out by categorising the variables into the different themes and ran codes to see which ones had most influence on Student Teaccher Ratios. Availability of high school, Primary Schools having toilets, Availability of Middle School, Availability of Midday meal scheme, Anganwadi centres, Senior secondary schools were the dominant variables under education. Internal pucca roads, piper water connectivity, recreational centres, post office and water availability, open covered drainage all have an important role to play in Tamil Nadu's STR. Total households availing pension, pregnant women, pregnant women receiving care, total number of lactating mothers, children aged 0 to 3 years, phcs, subcentres were factors falling under health. We also found railway station availability, children not attending school, households with clean energy, availability of early education centre to all influence STR.

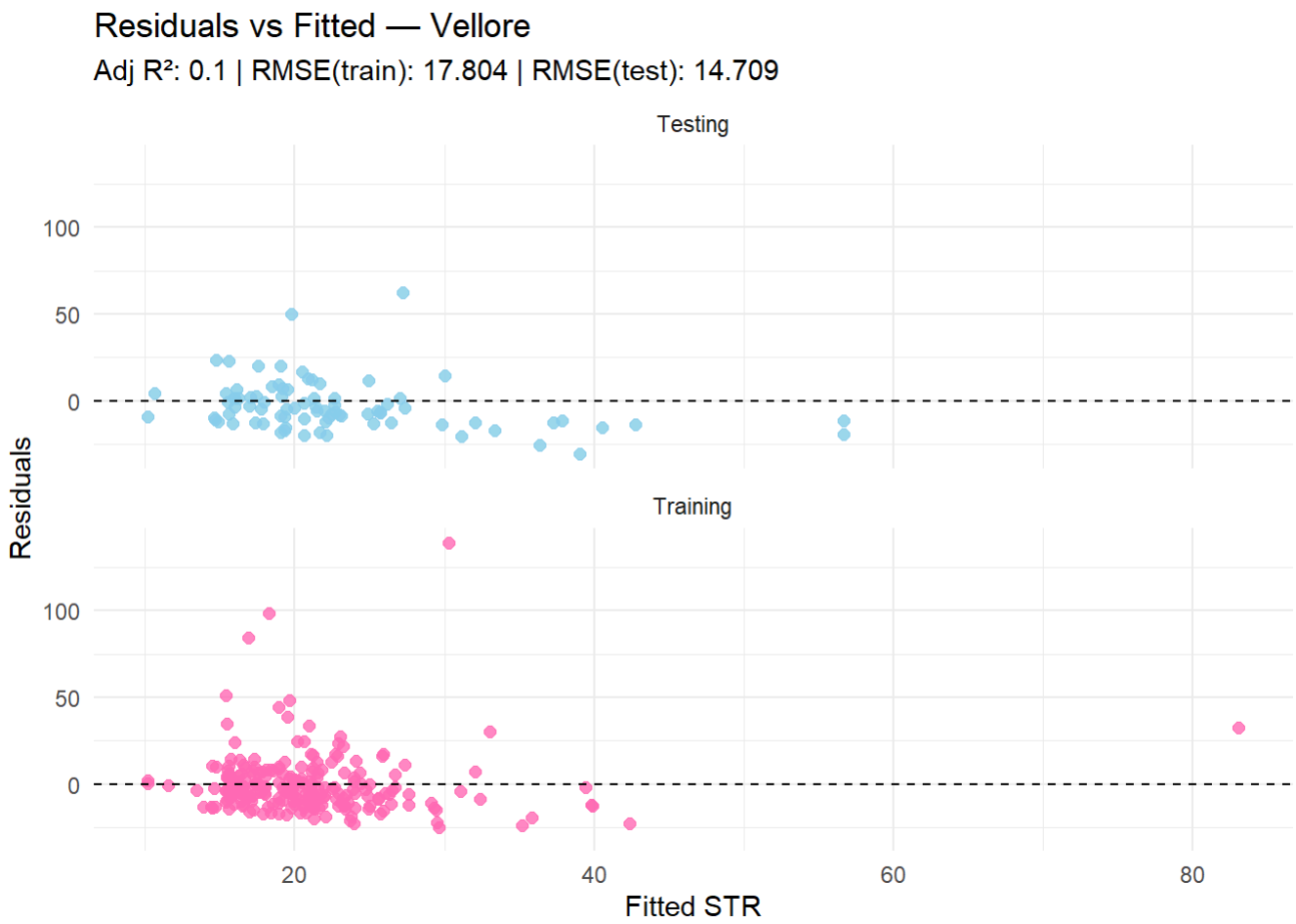
```
## Processing district: thiruvallur
```



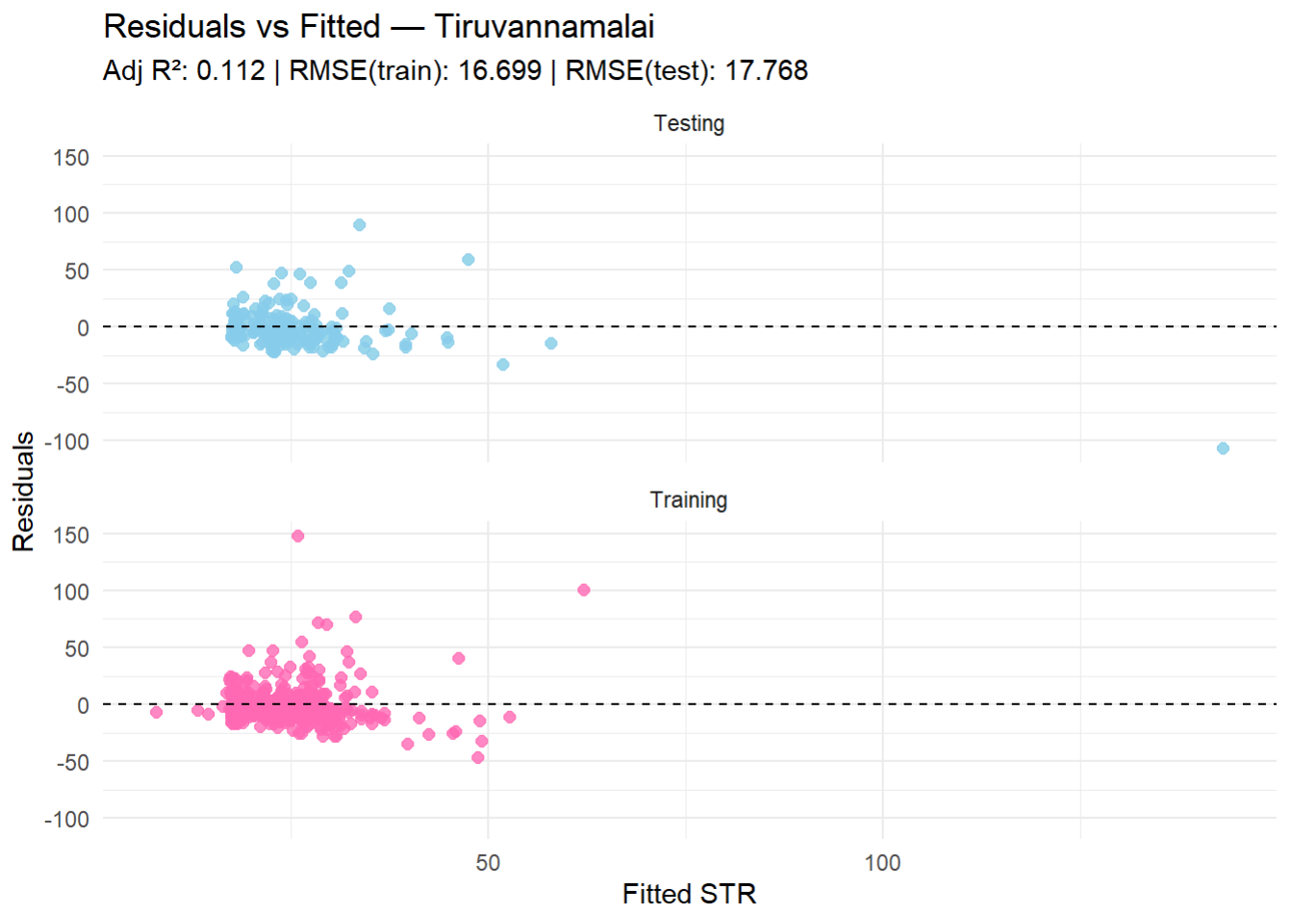

```
## Processing district: kancheepuram
```



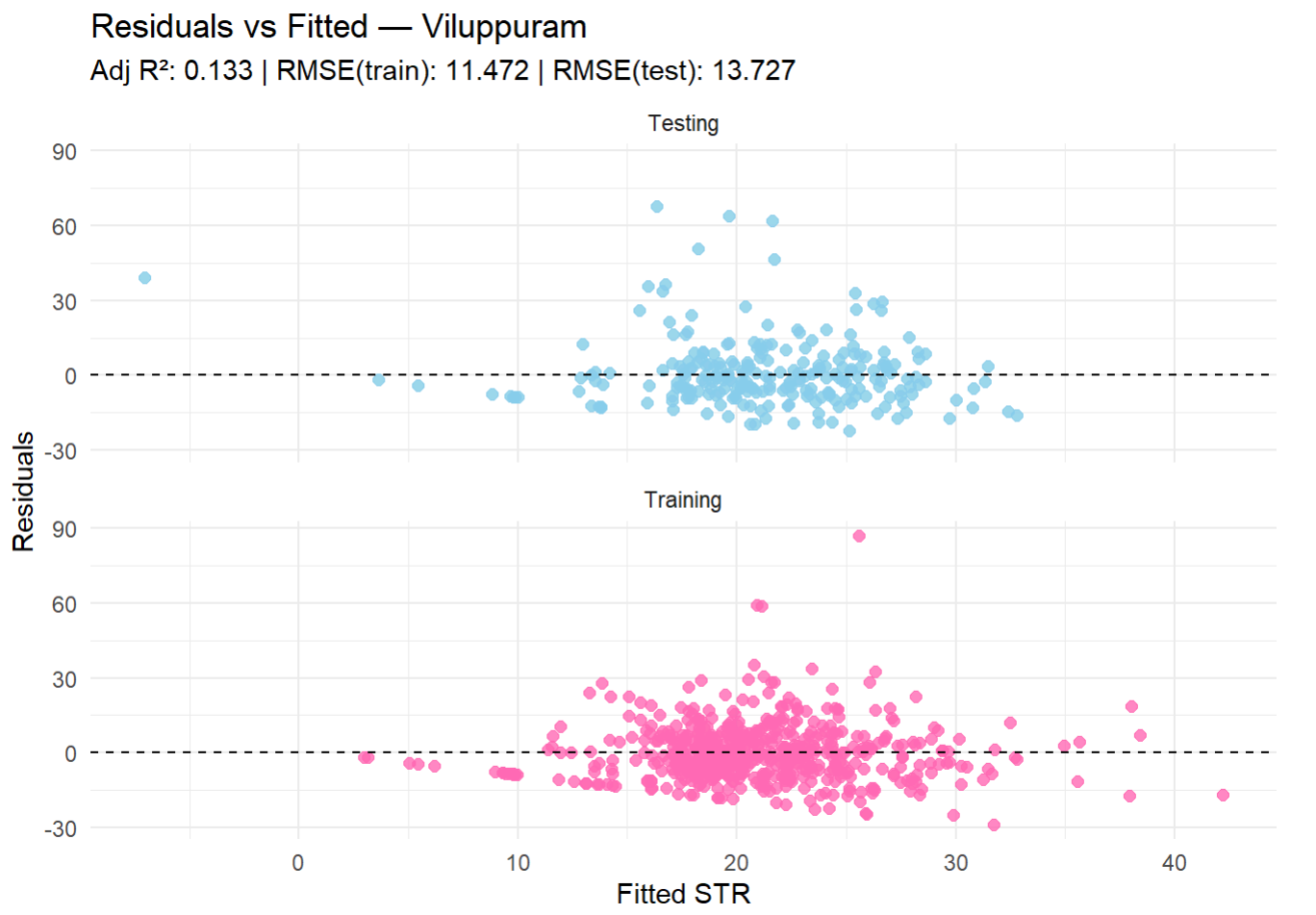
```
## Processing district: vellore
```



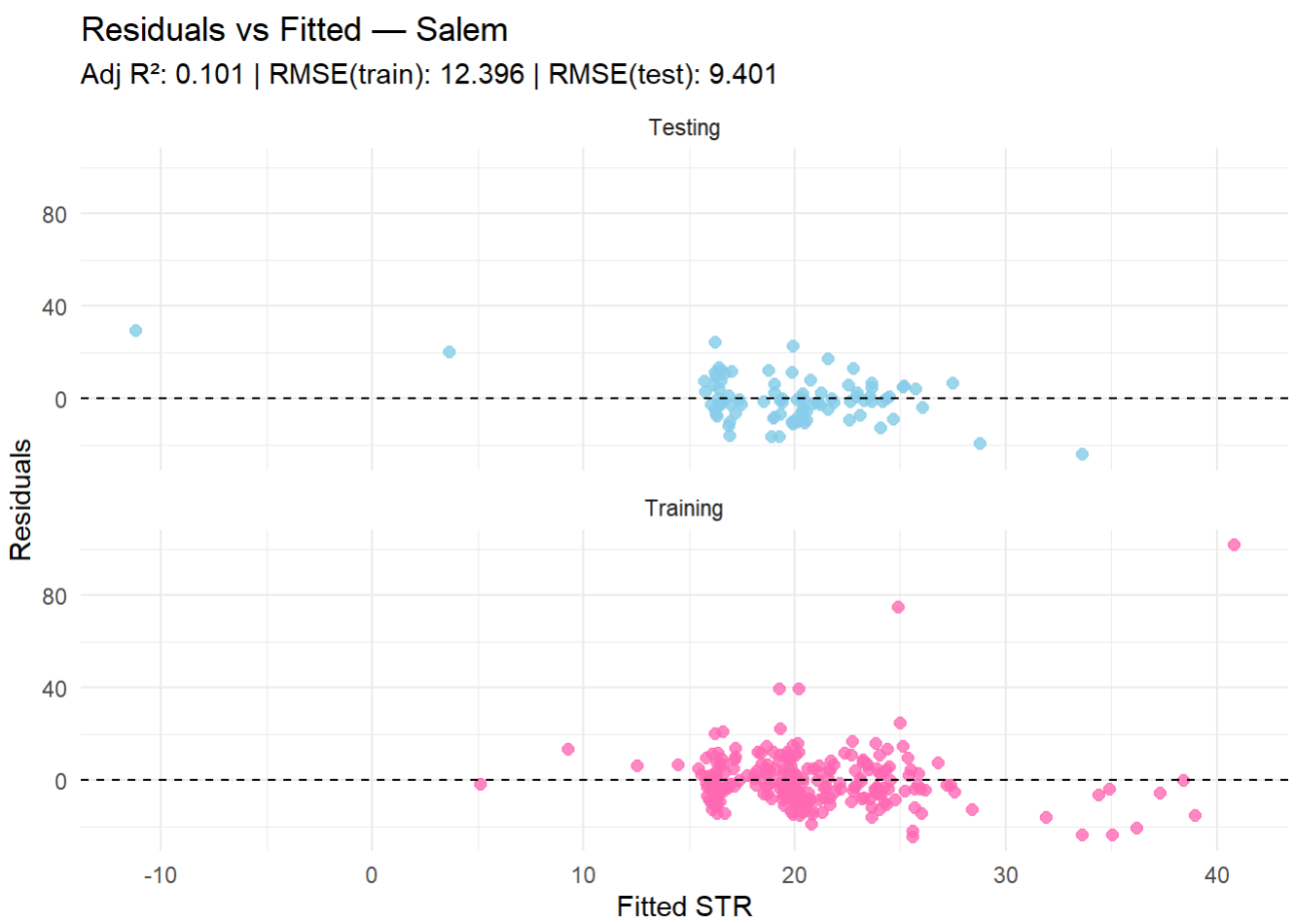

```
## Processing district: tiruvannamalai
```



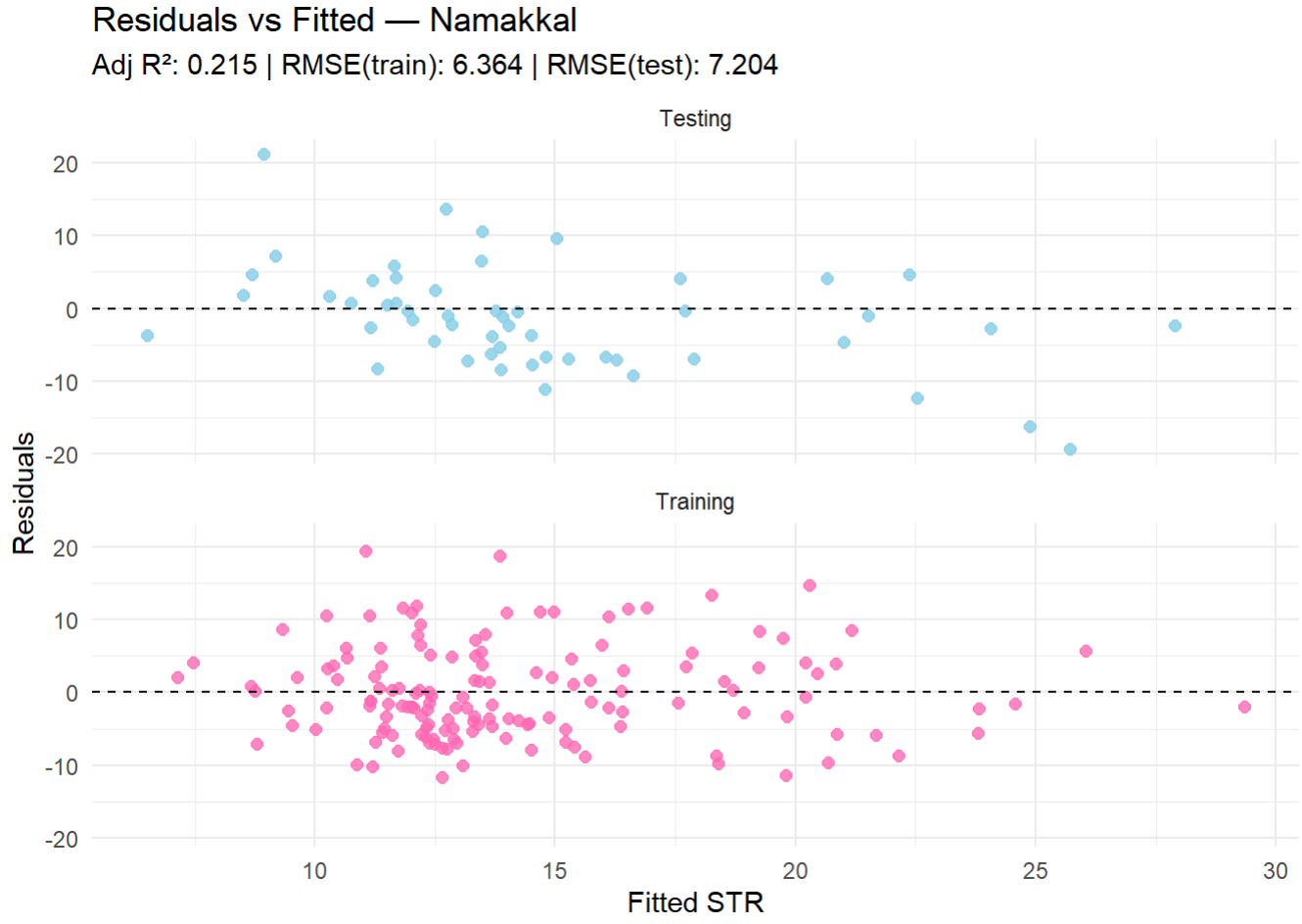

```
## Processing district: viluppuram
```



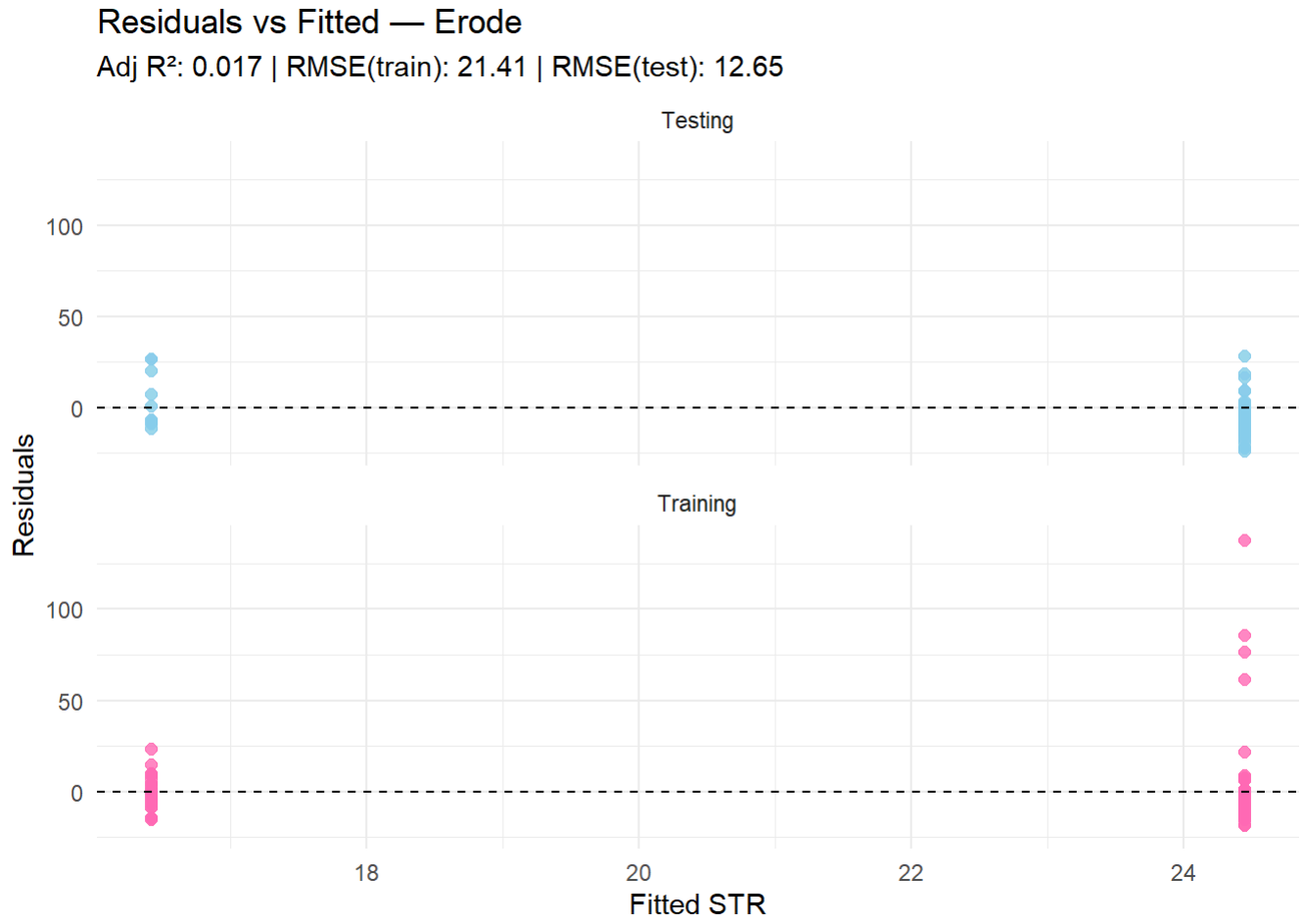
```
## Processing district: salem
```




```
## Processing district: namakkal
```

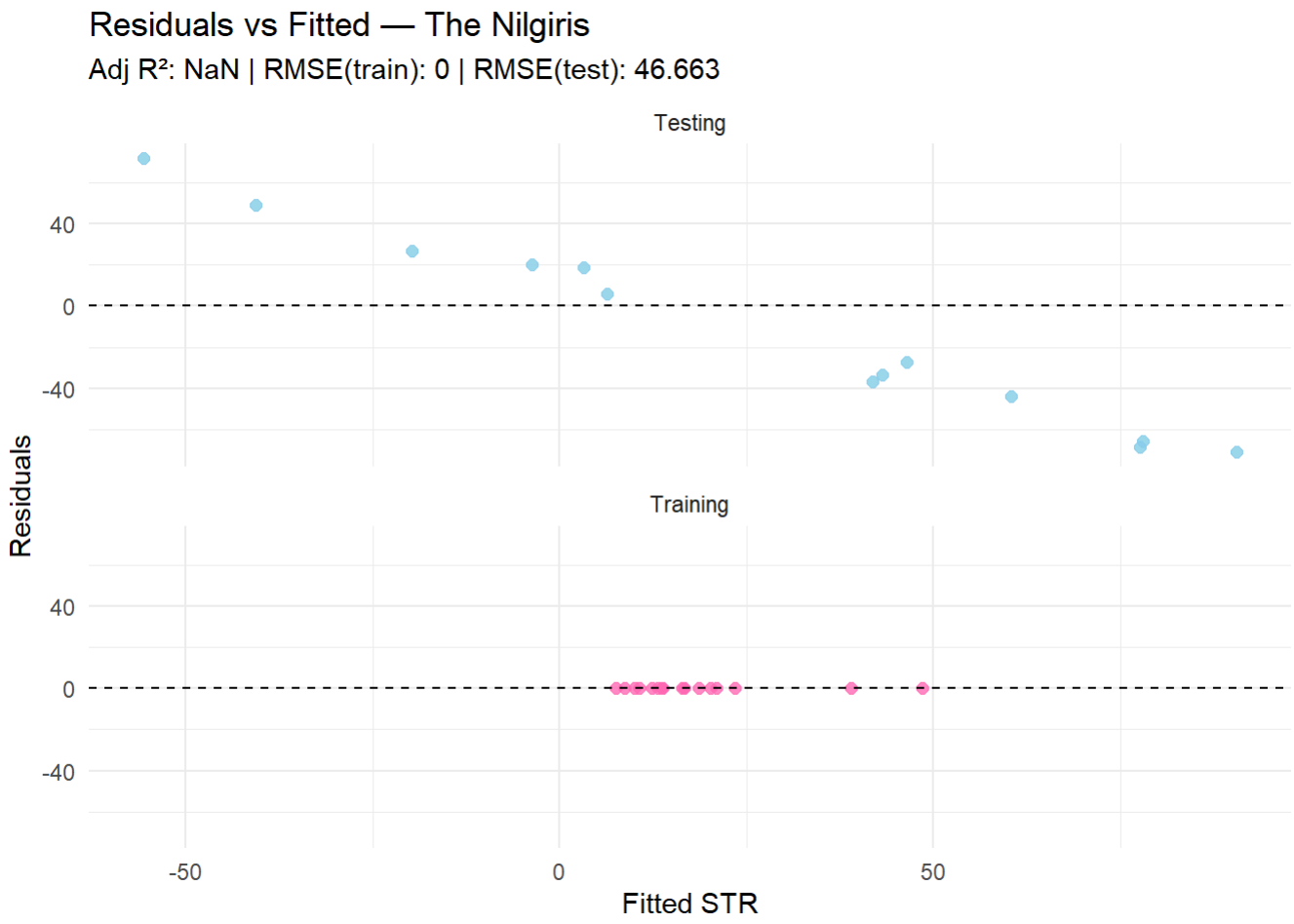



```
## Processing district: erode
```

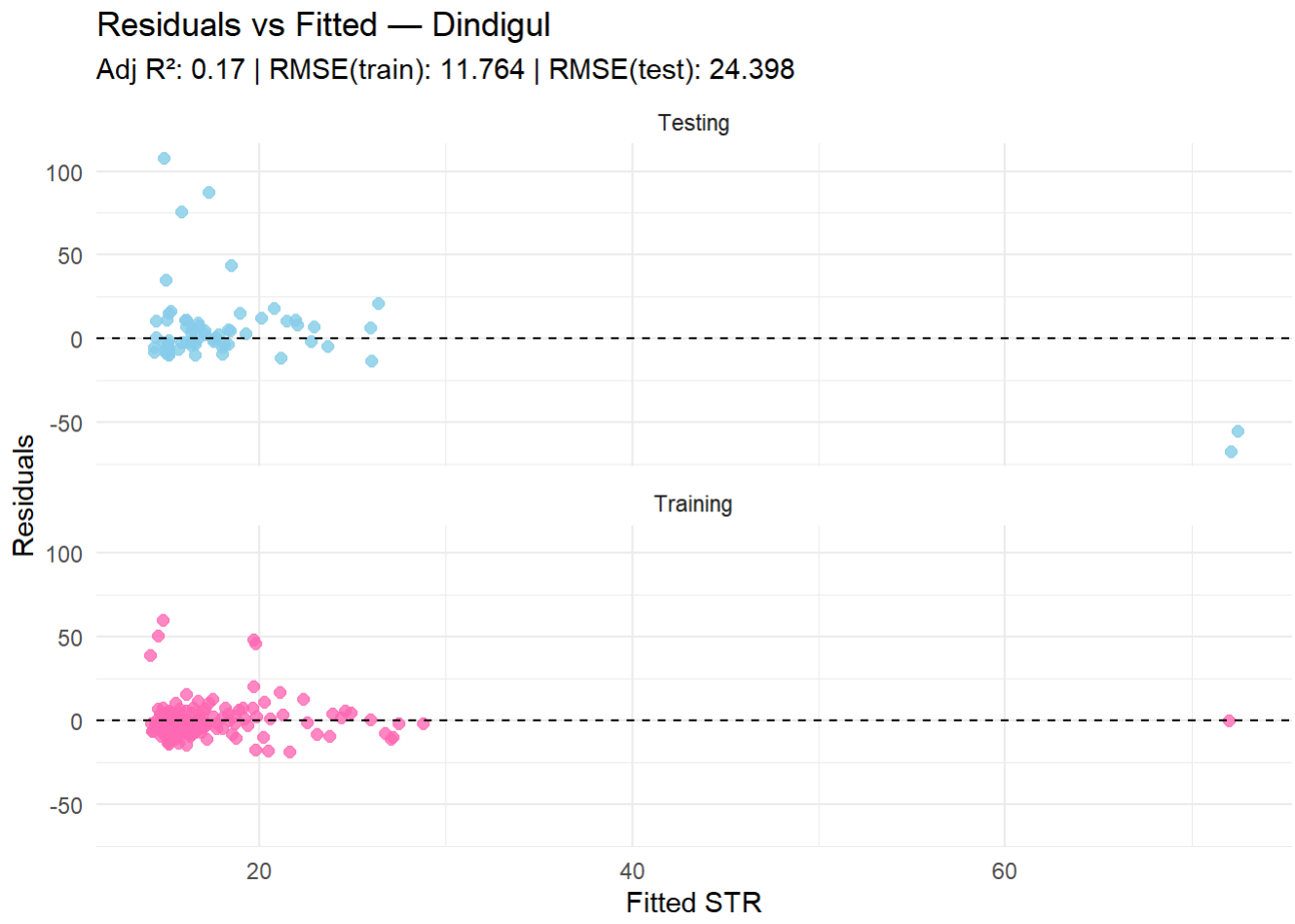


```
## Processing district: the nilgiris
```

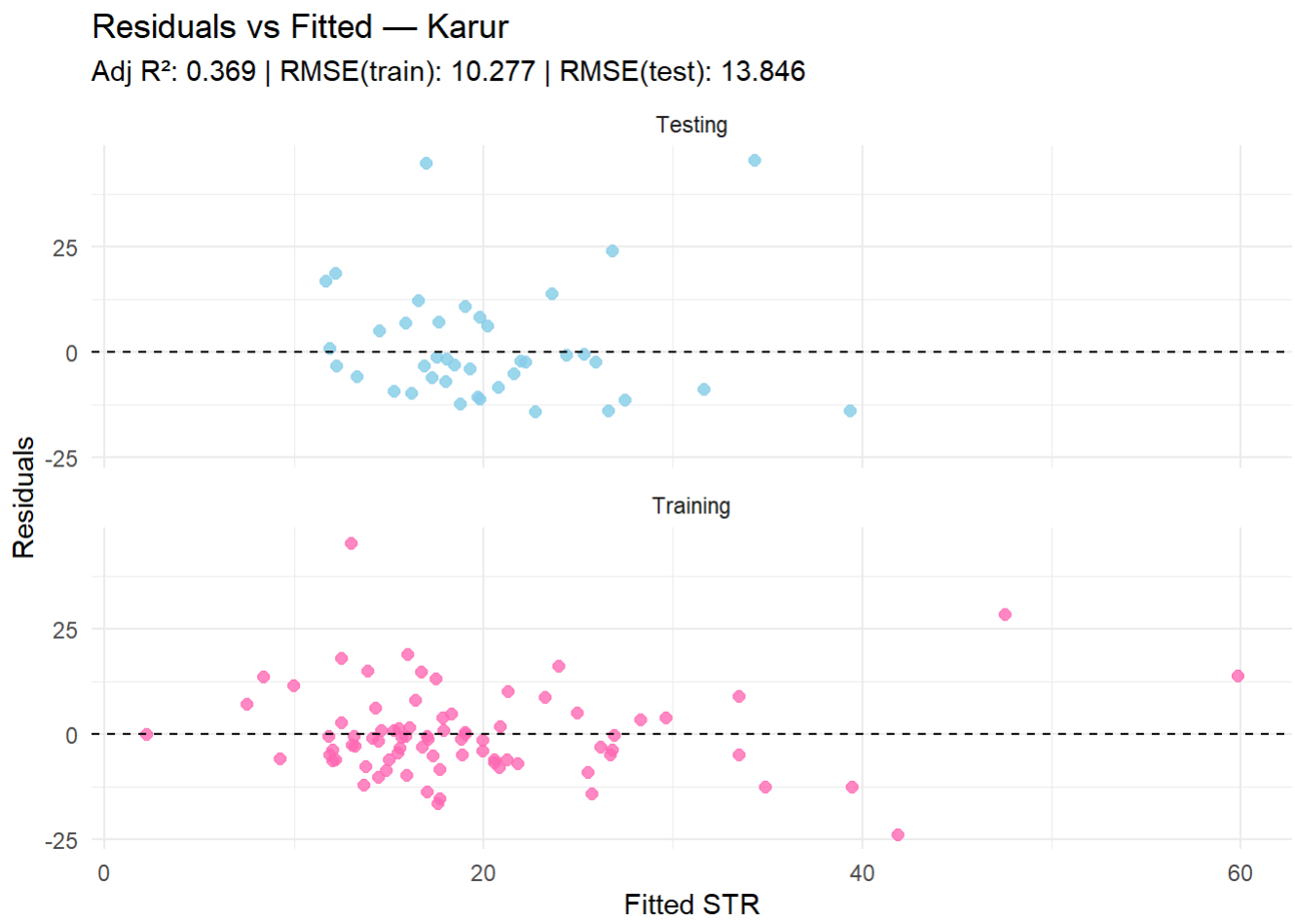
```
## Warning in predict.lm(model_step, newdata = test): prediction from
## rank-deficient fit; attr(*, "non-estim") has doubtful cases
```



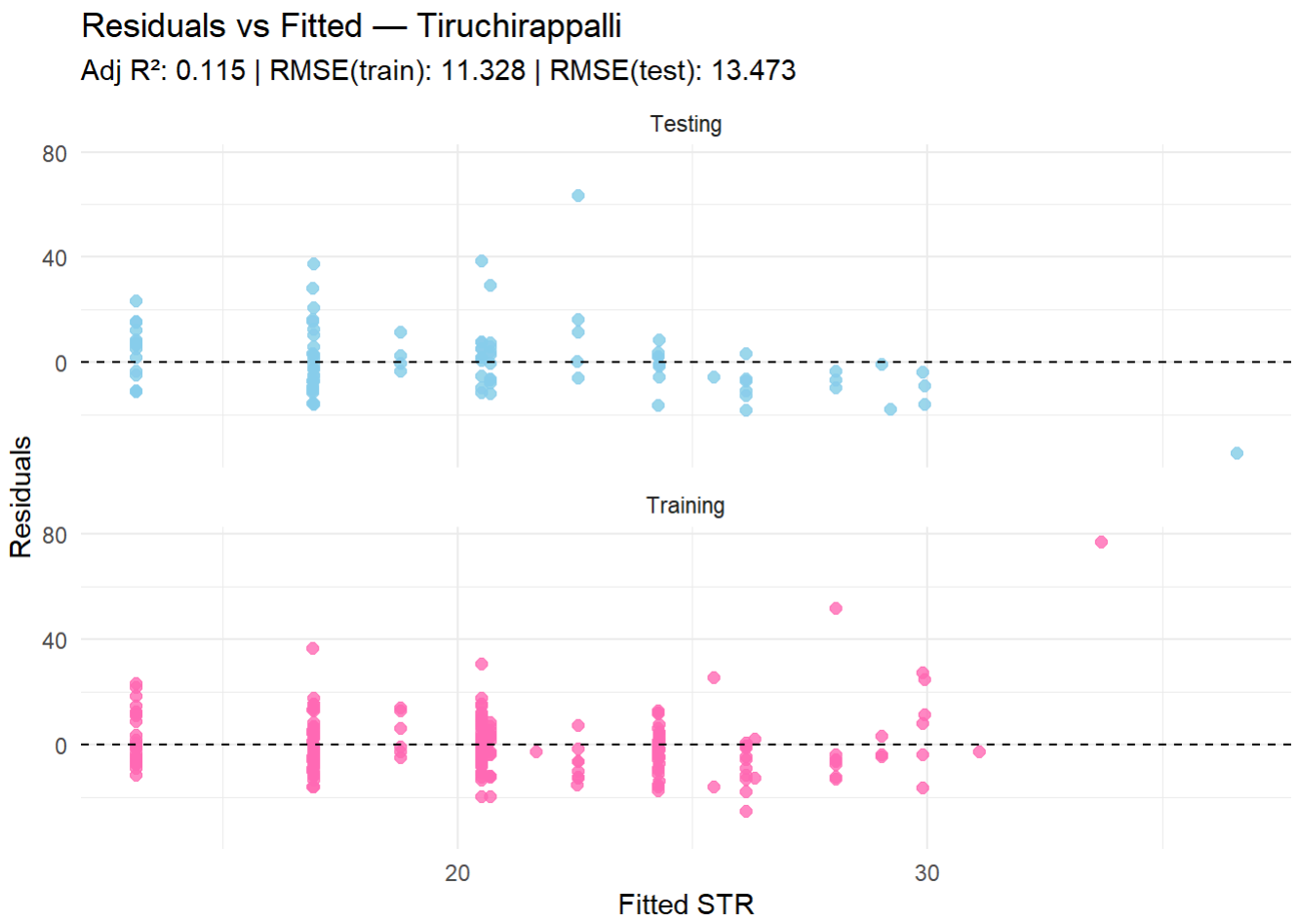

```
## Processing district: dindigul
```



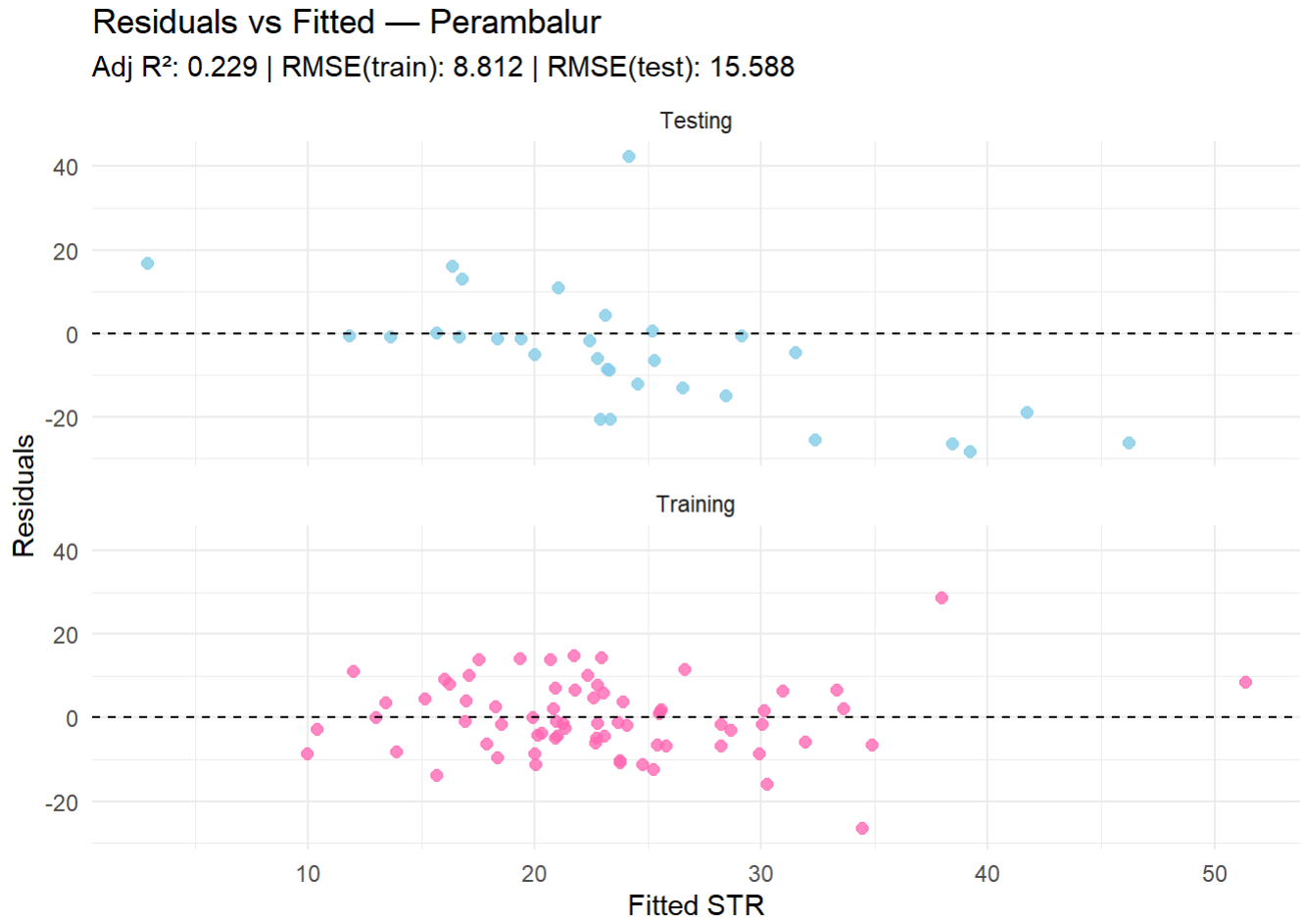

```
## Processing district: karur
```



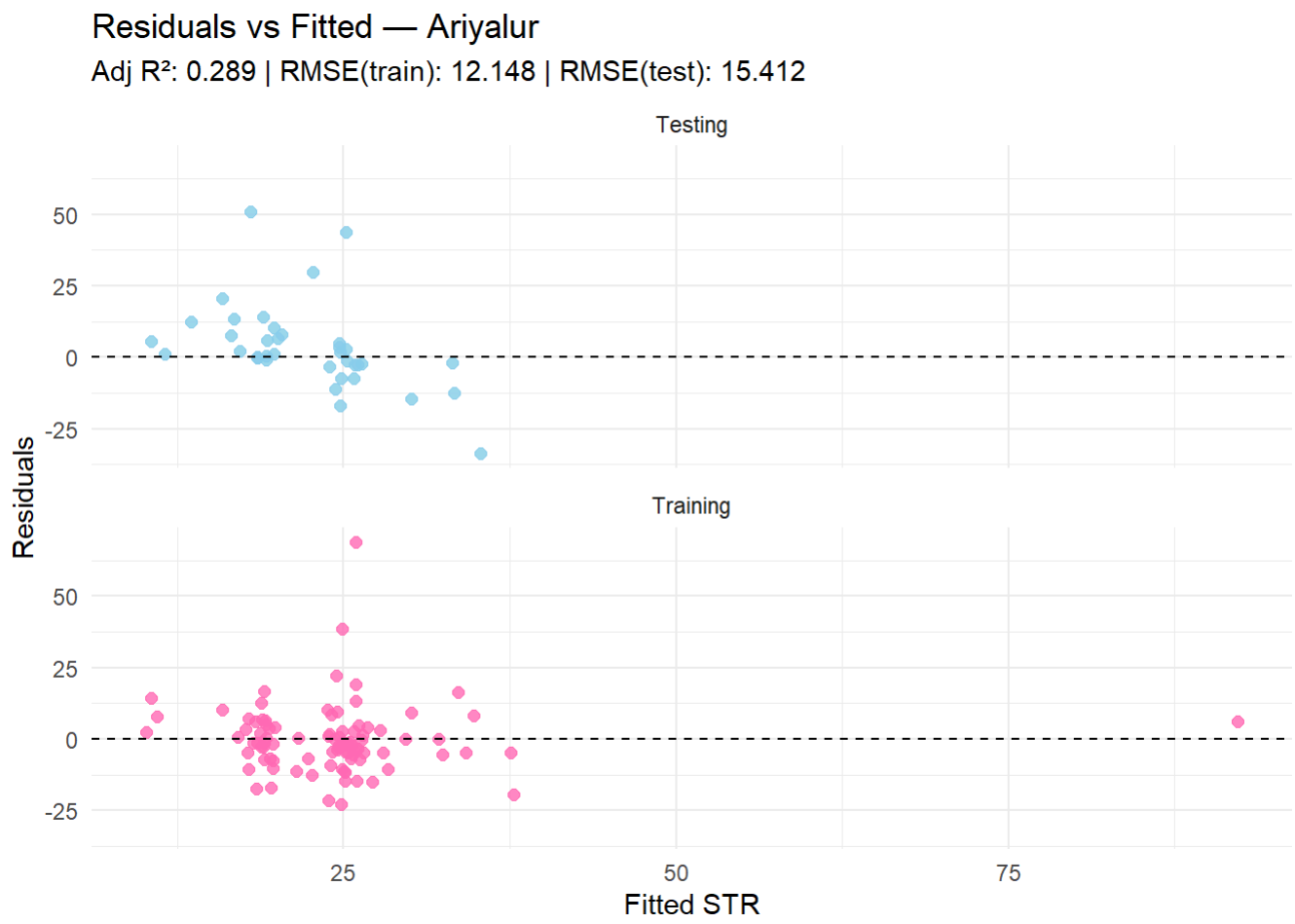
```
## Processing district: tiruchirappalli
```




```
## Processing district: perambalur
```

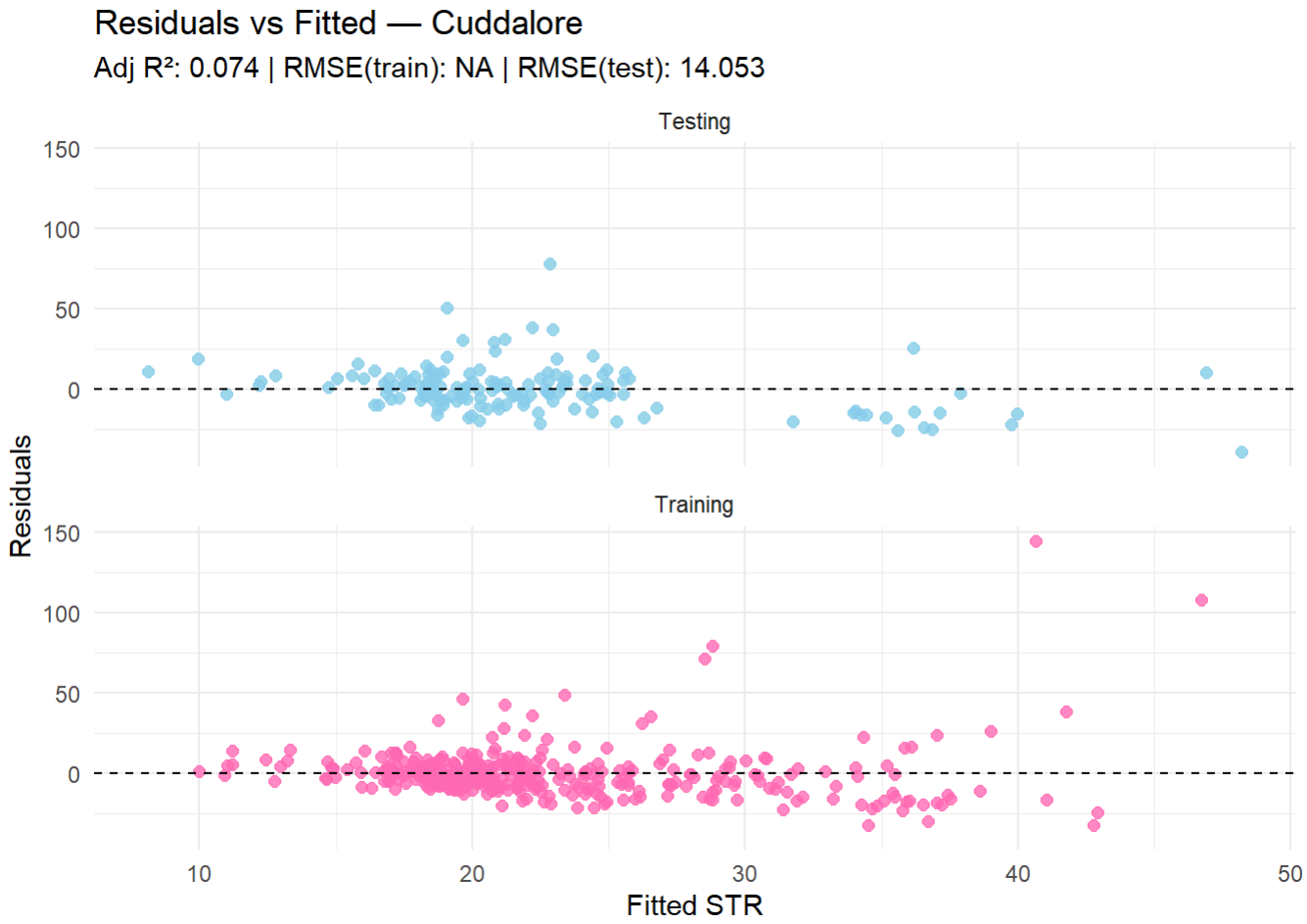



```
## Processing district: ariyalur
```

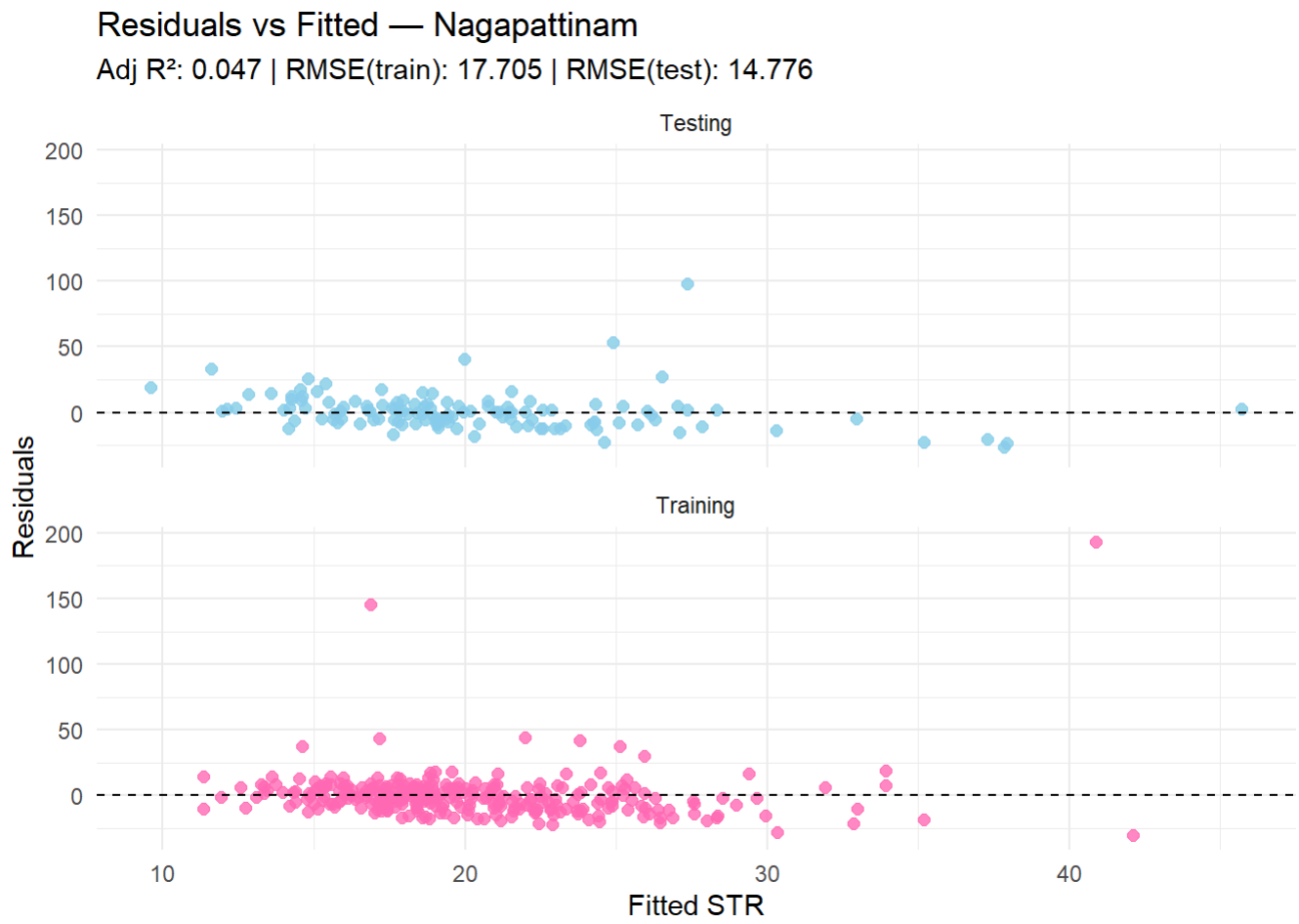


```
## Processing district: cuddalore
```

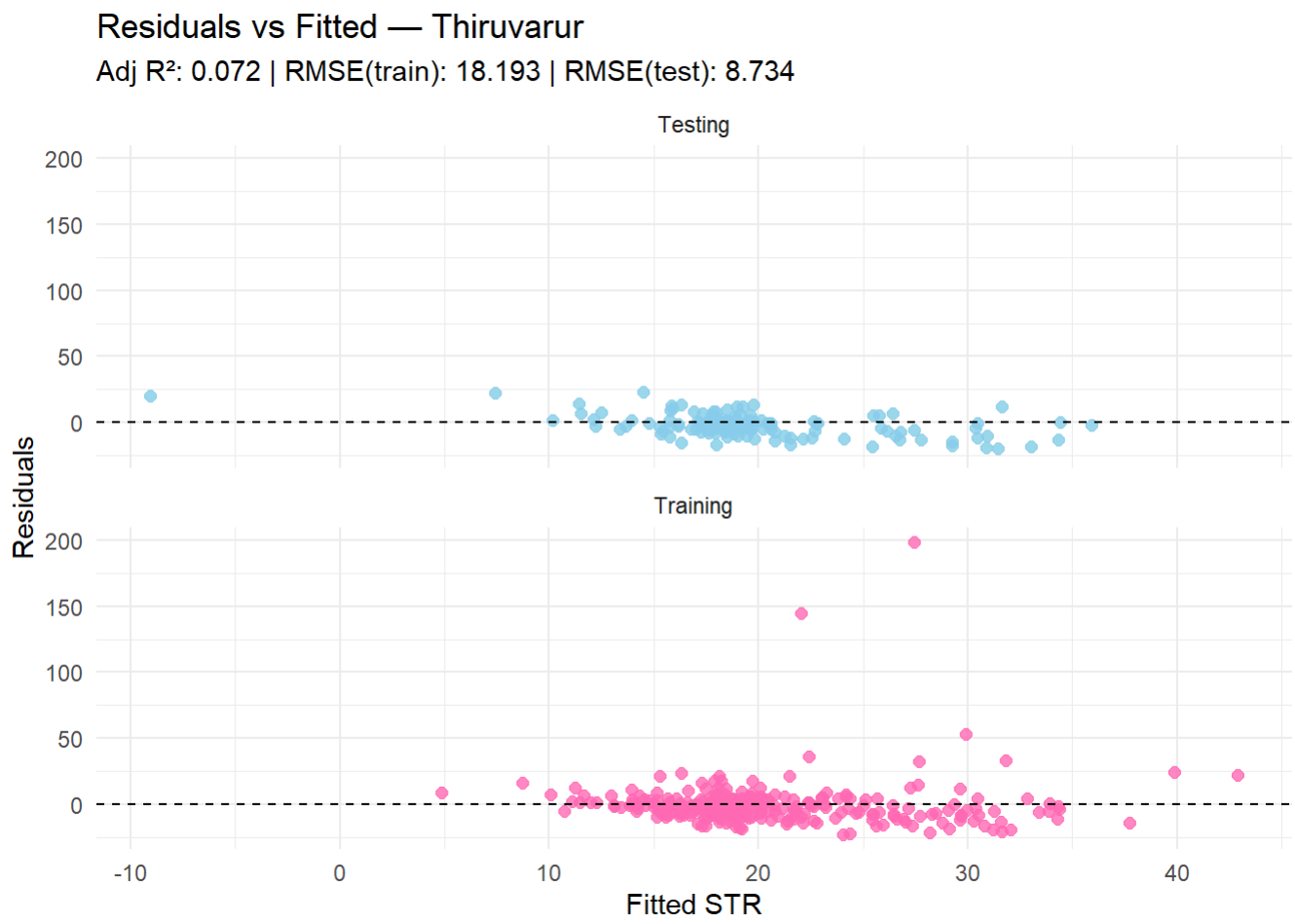
```
## Warning in is.na(actual) | is.na(pred): longer object length is not a multiple
## of shorter object length
```



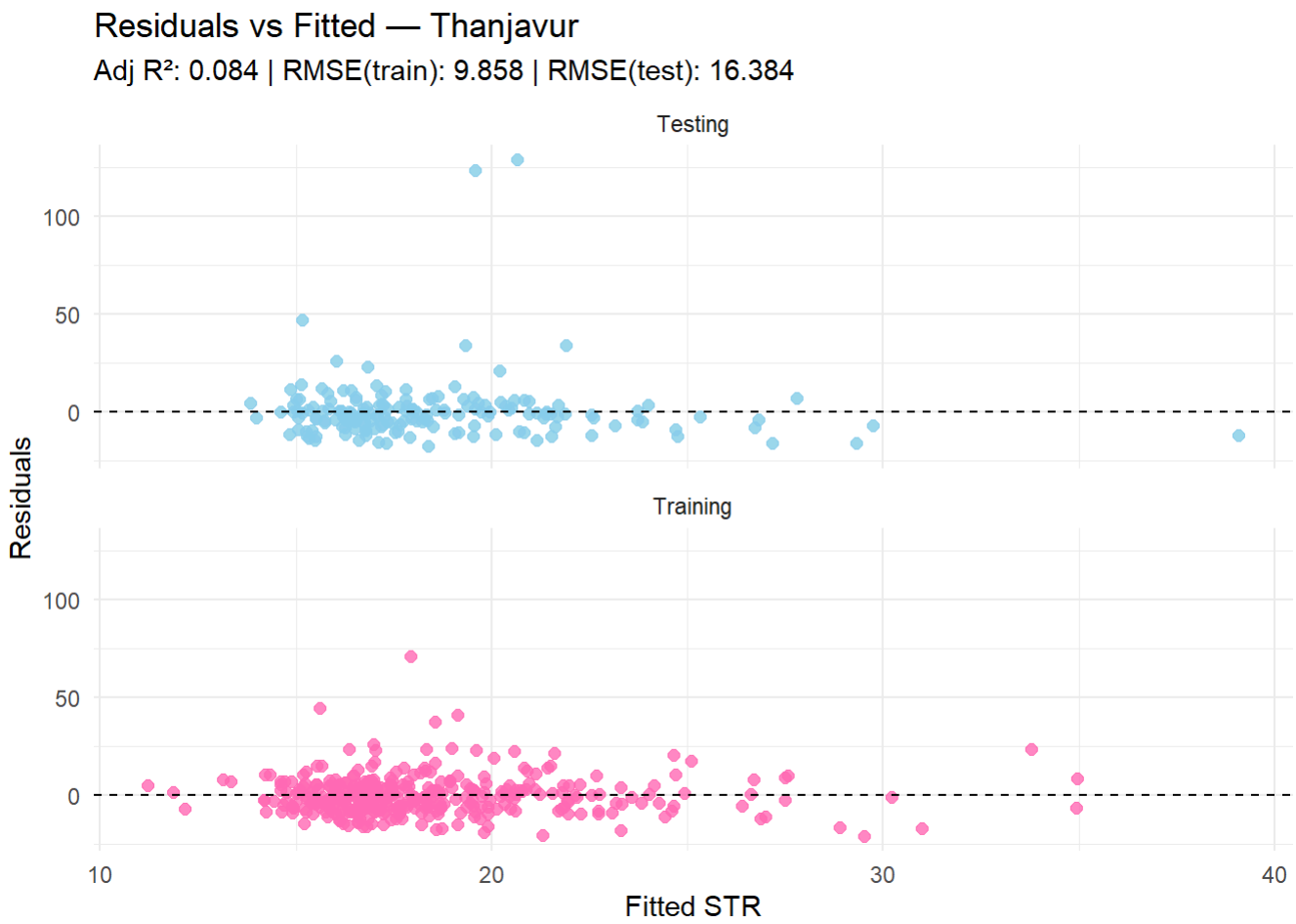

```
## Processing district: nagapattinam
```



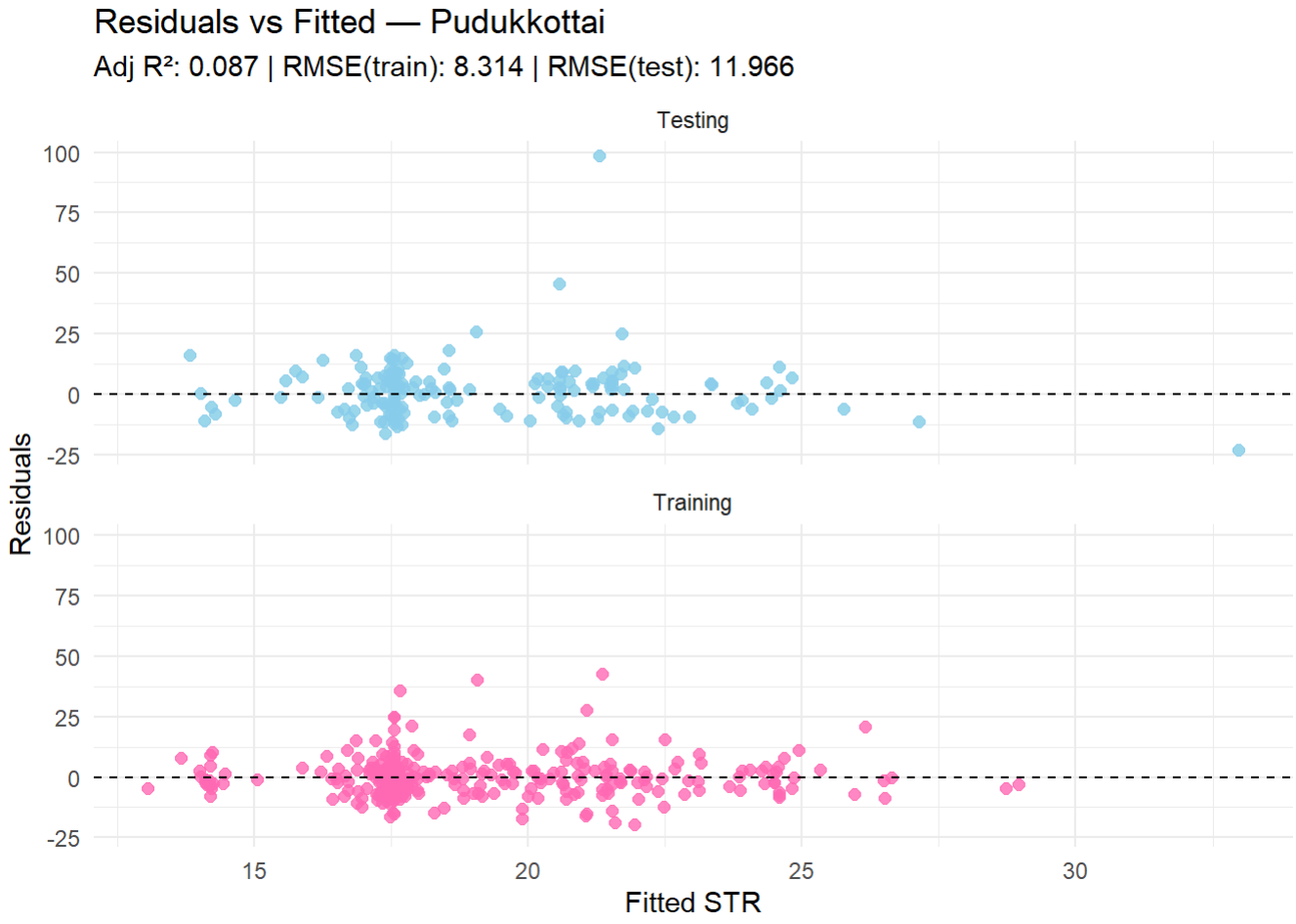

```
## Processing district: thiruvarur
```



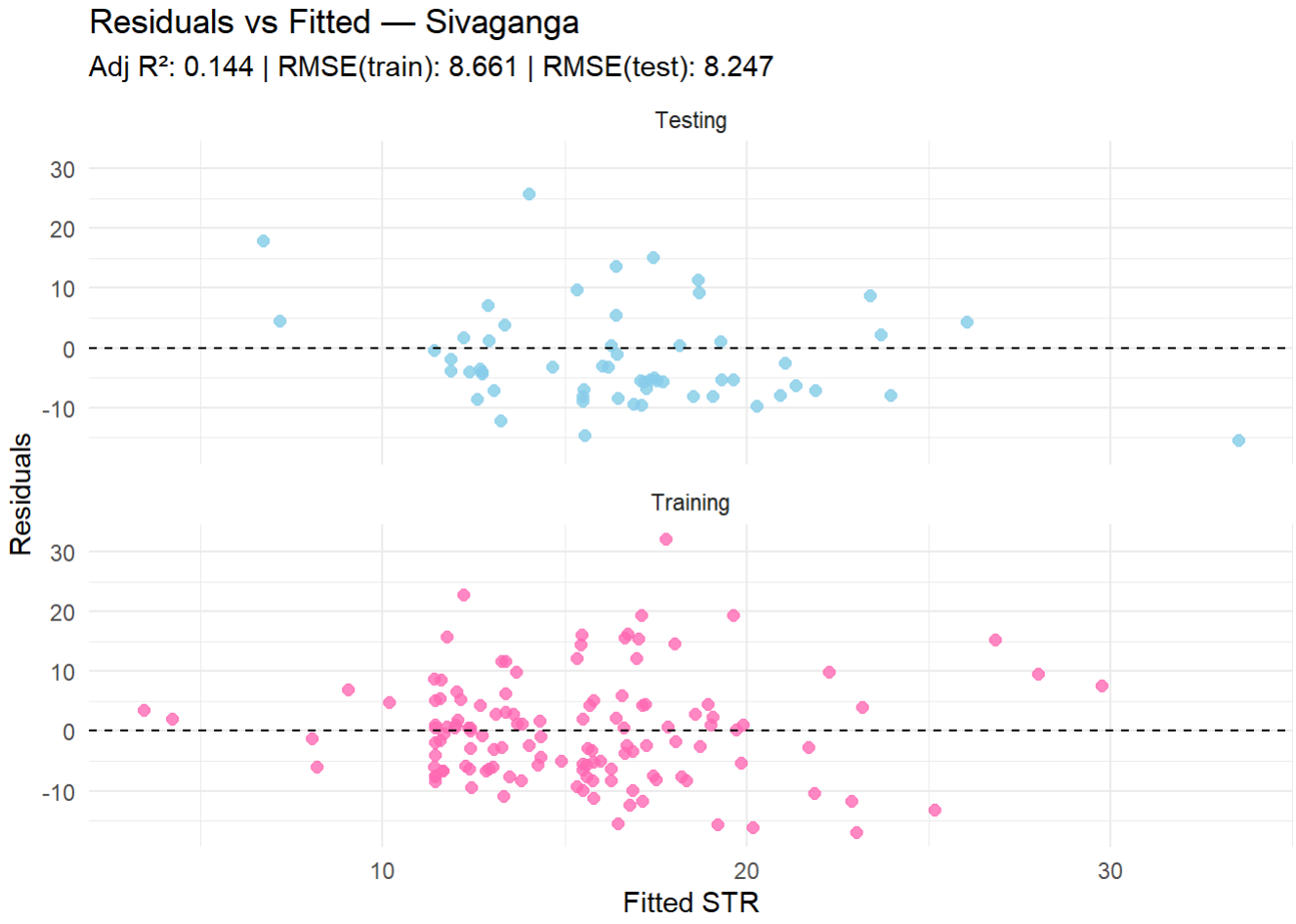
```
## Processing district: thanjavur
```



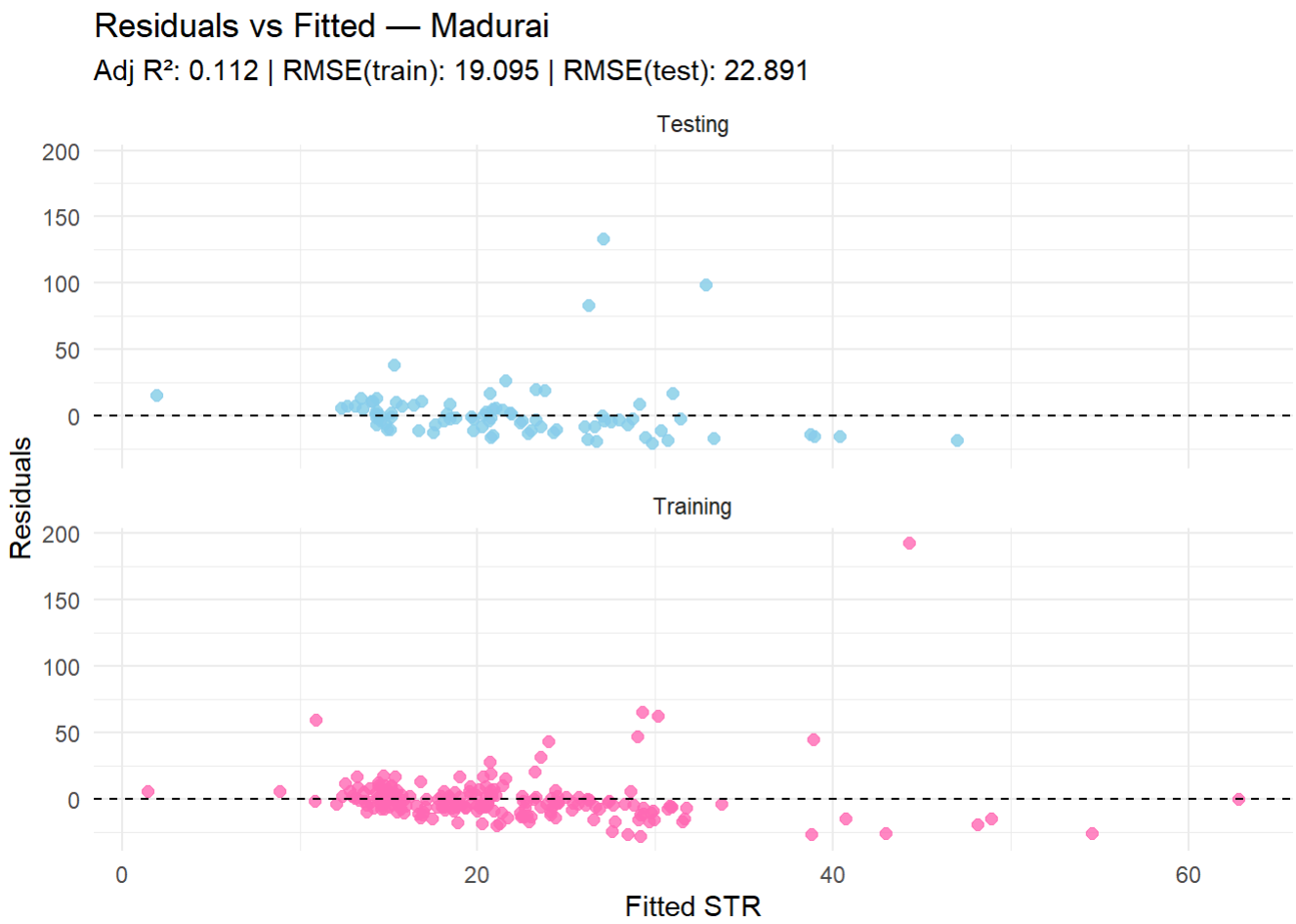

```
## Processing district: pudukkottai
```



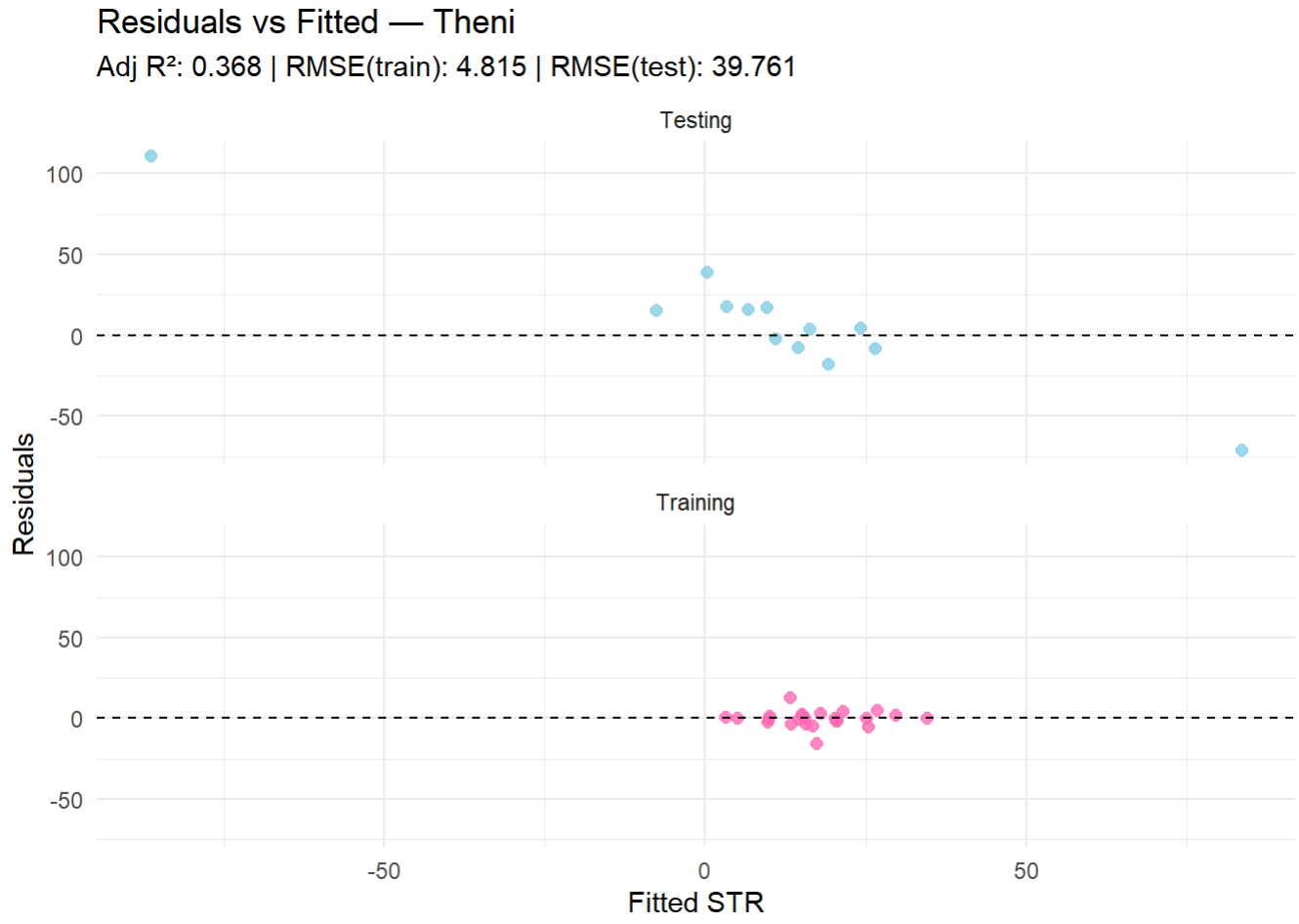

```
## Processing district: sivaganga
```



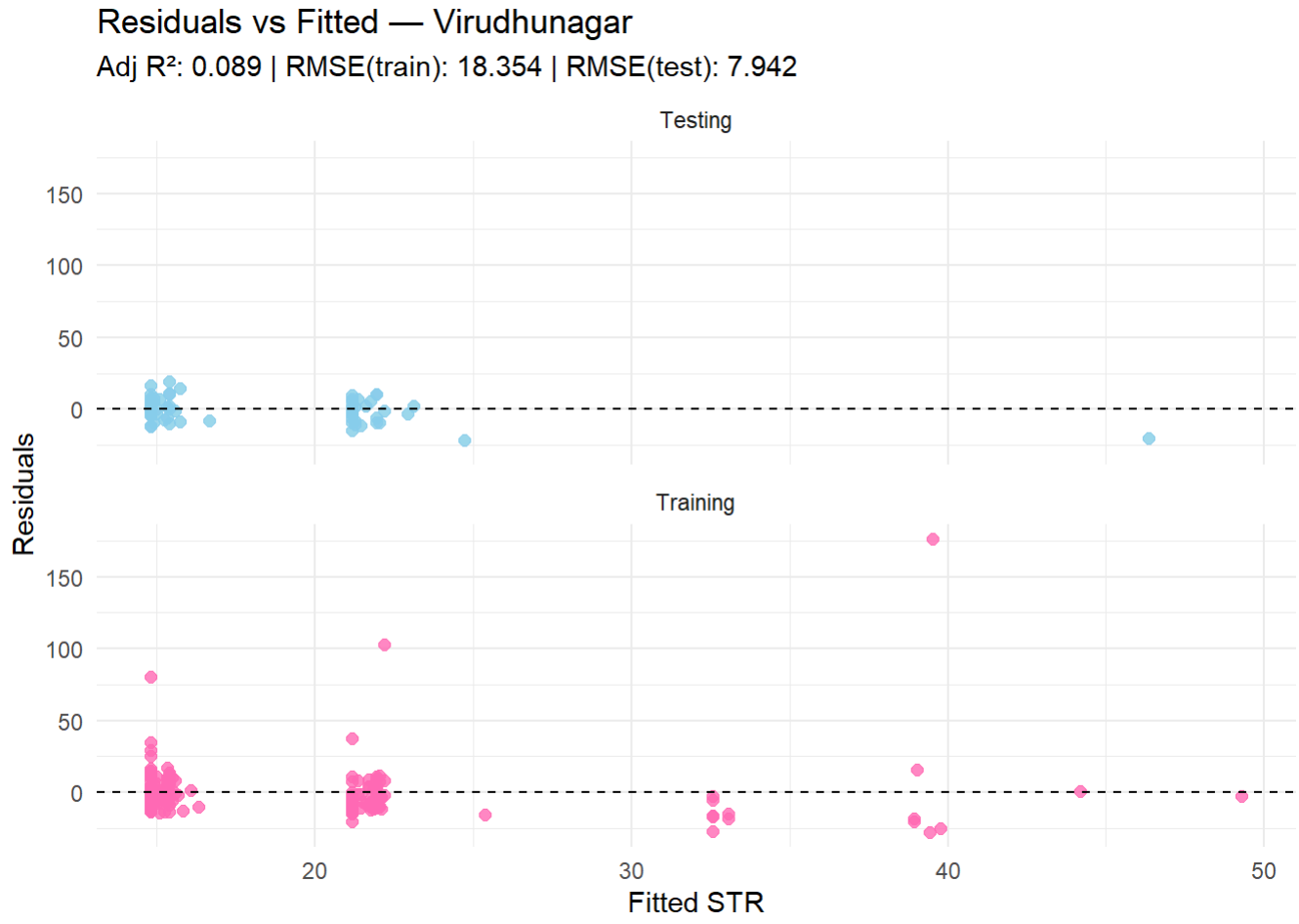
```
## Processing district: madurai
```



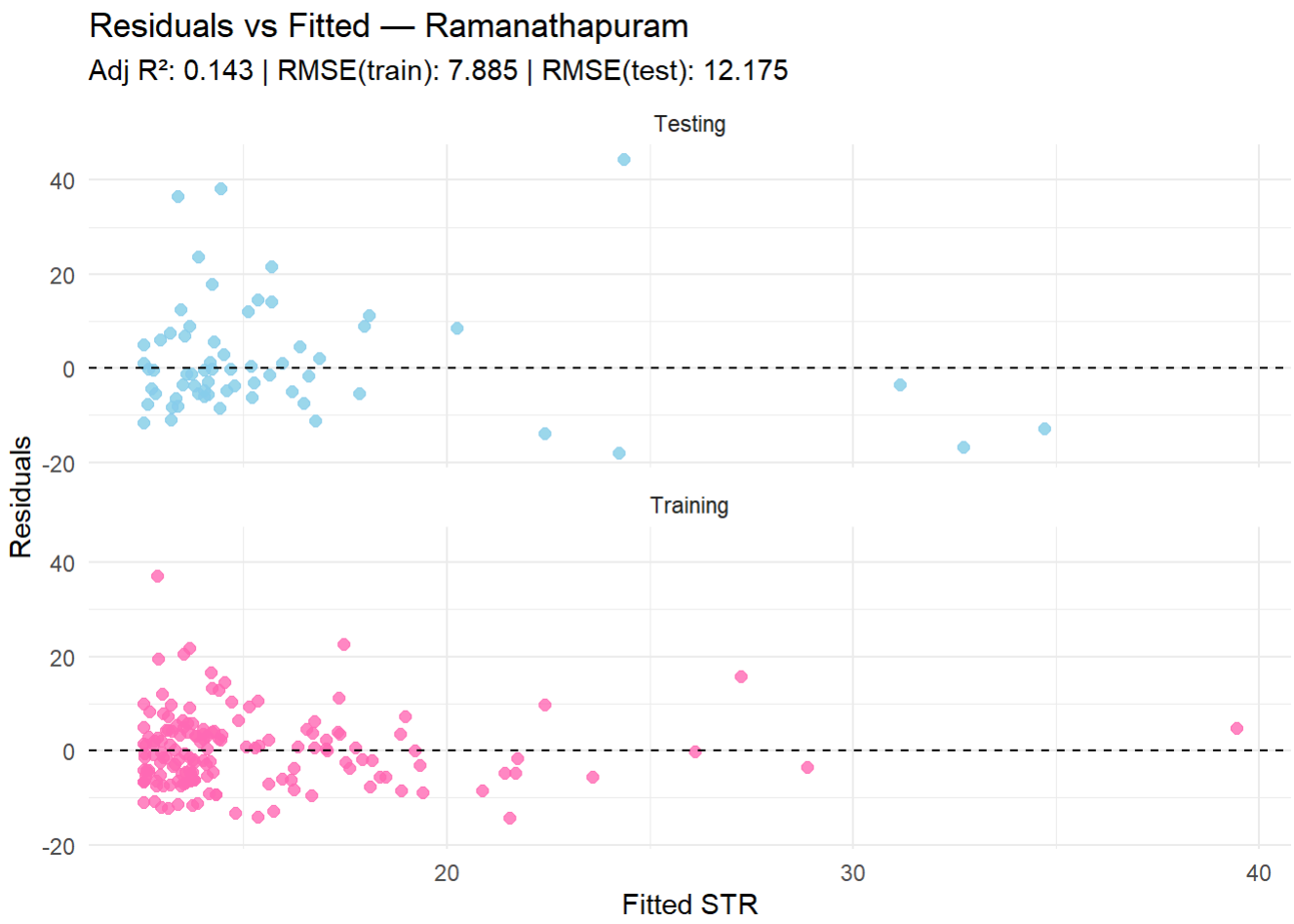

```
## Processing district: theni
```



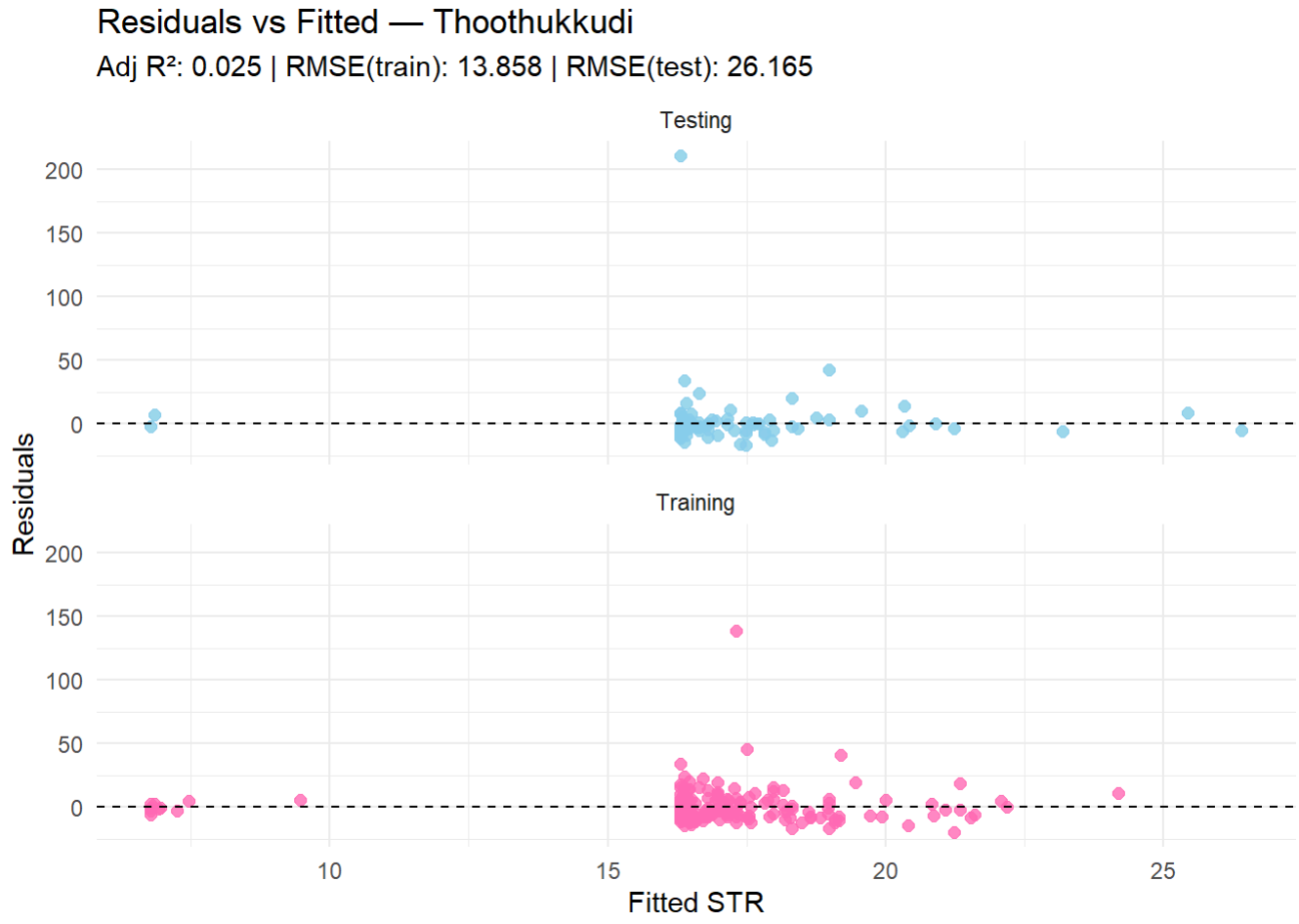

```
## Processing district: virudhunagar
```



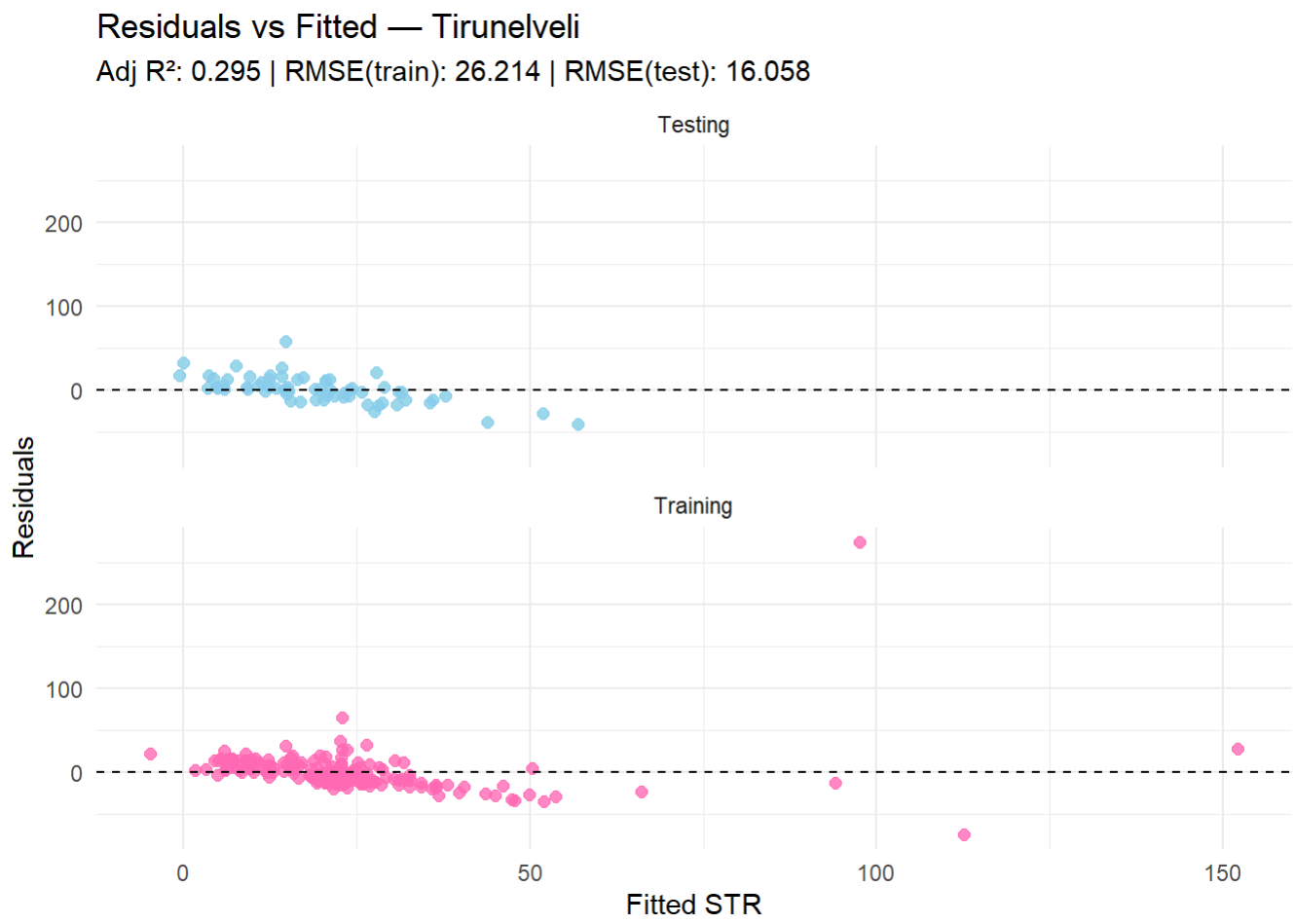
```
## Processing district: ramanathapuram
```




```
## Processing district: thoothukkudi
```

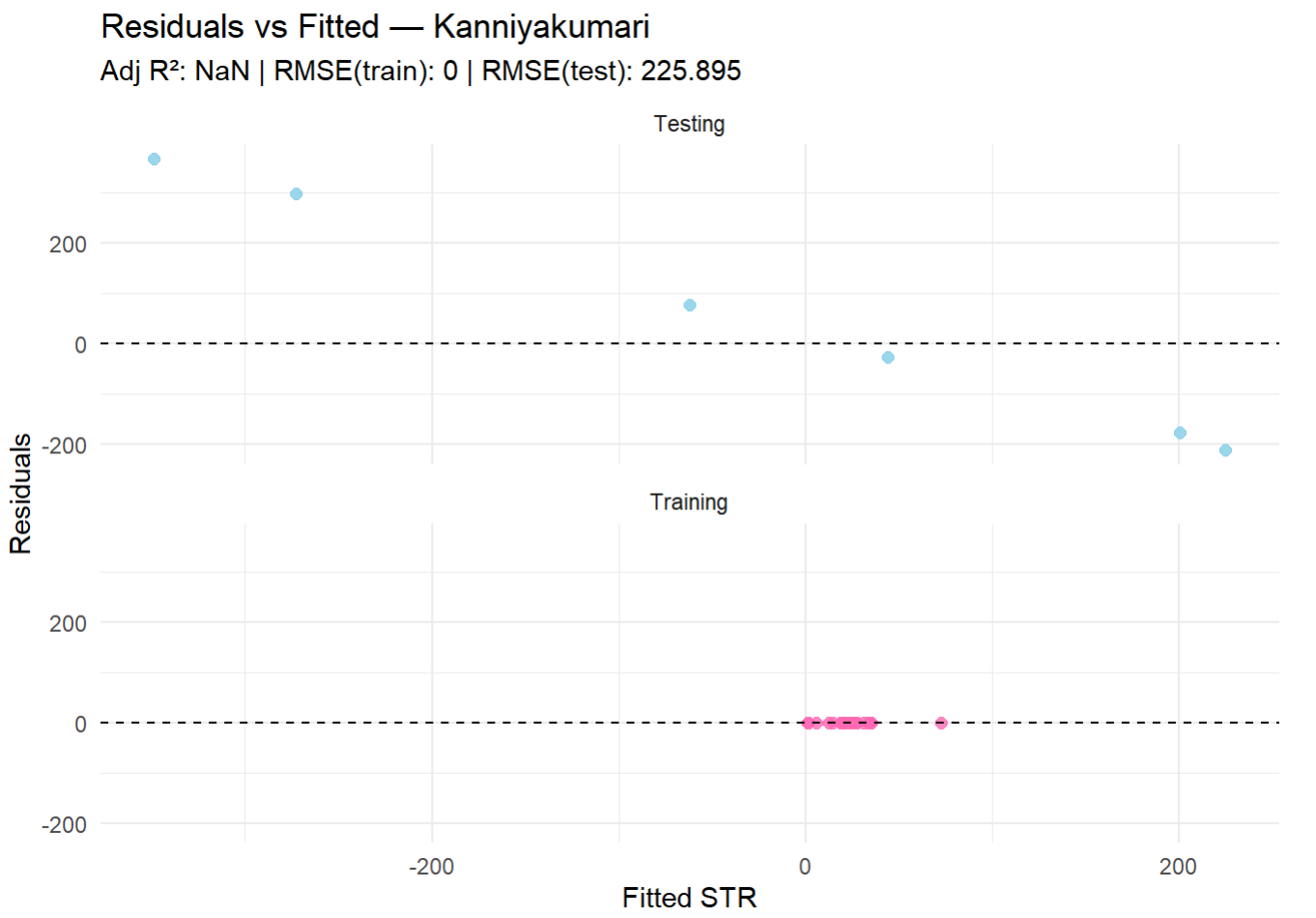



```
## Processing district: tirunelveli
```

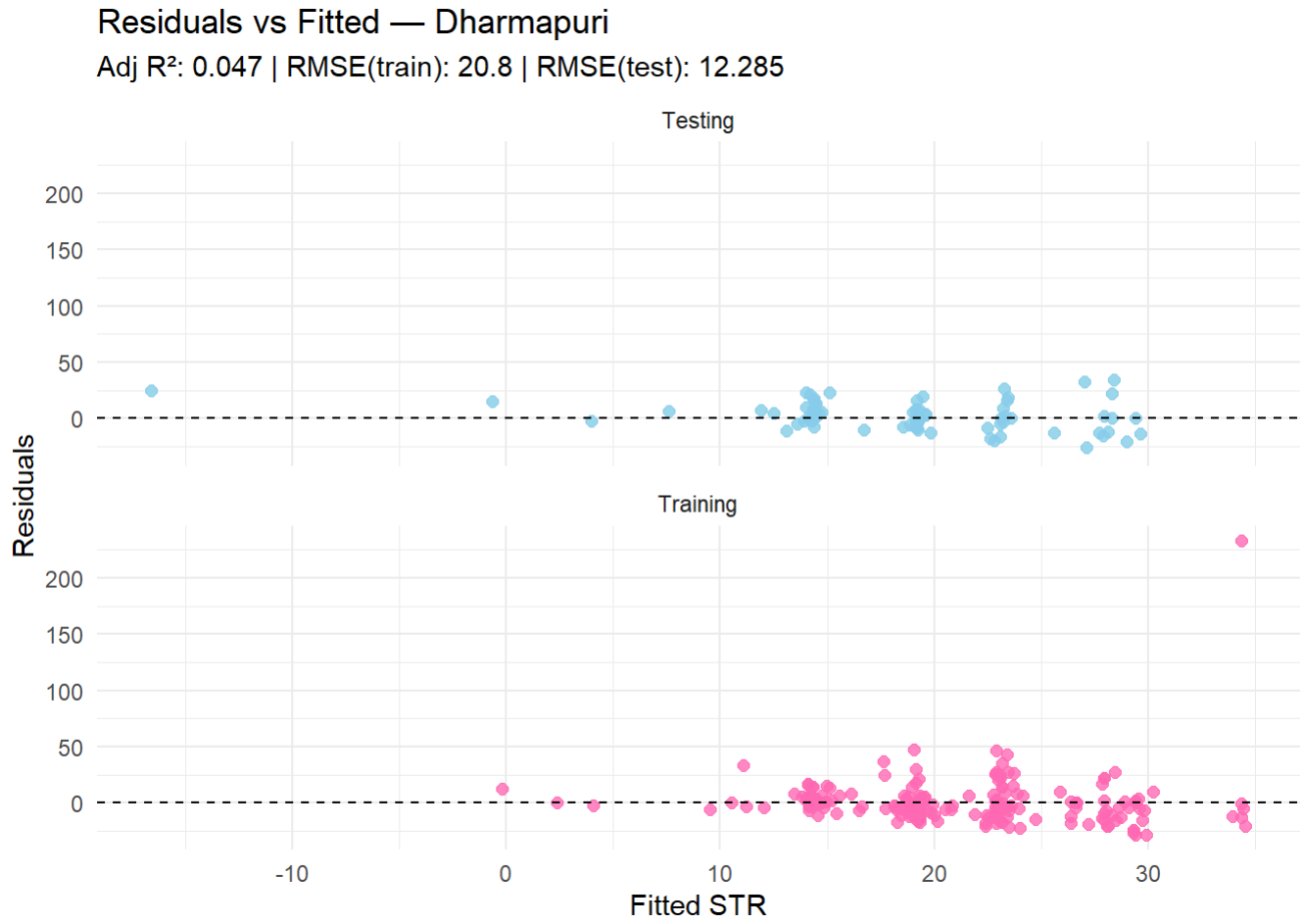


```
## Processing district: kanniyakumari
```

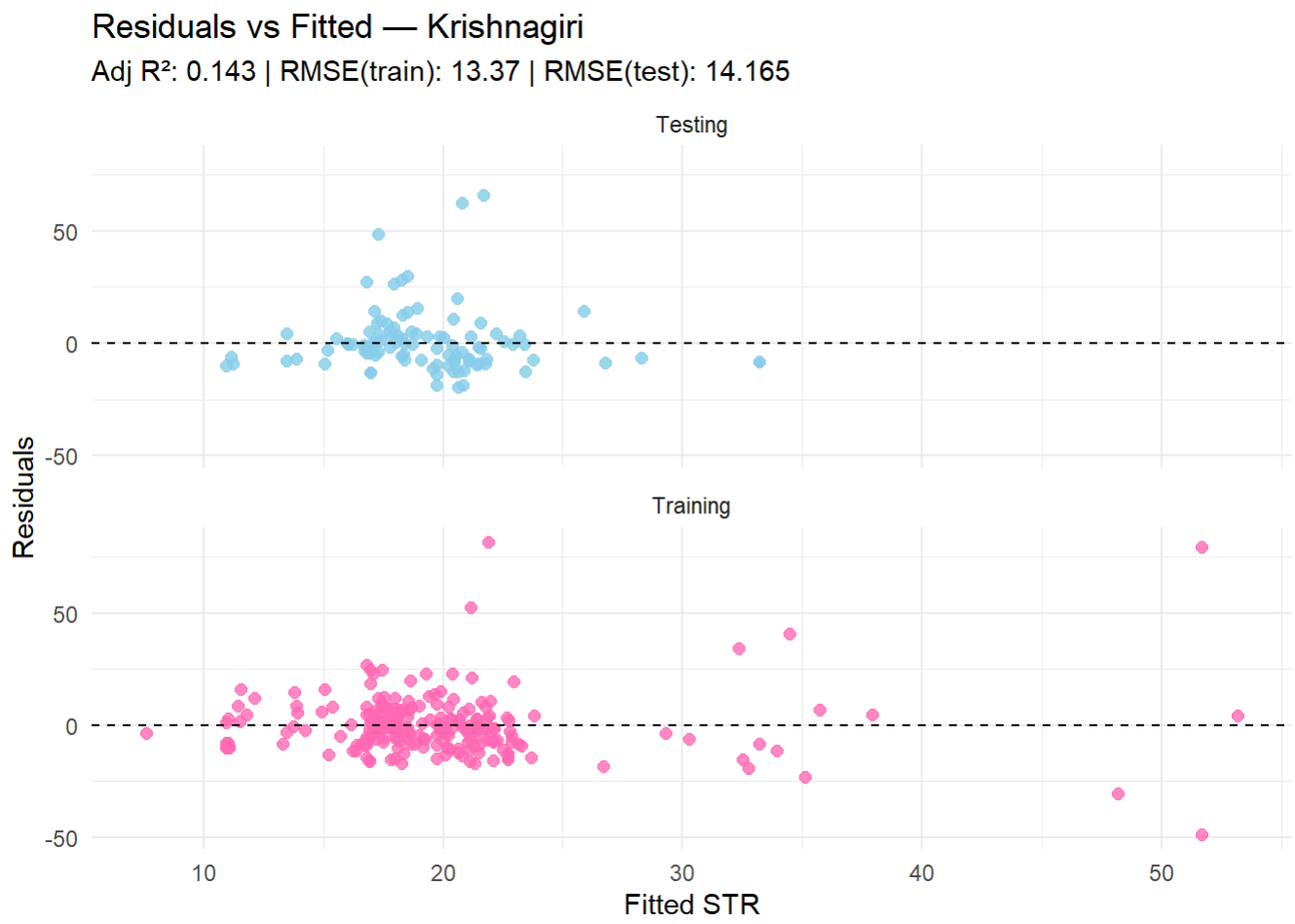
```
## Warning in predict.lm(model_step, newdata = test): prediction from
## rank-deficient fit; attr(*, "non-estim") has doubtful cases
```



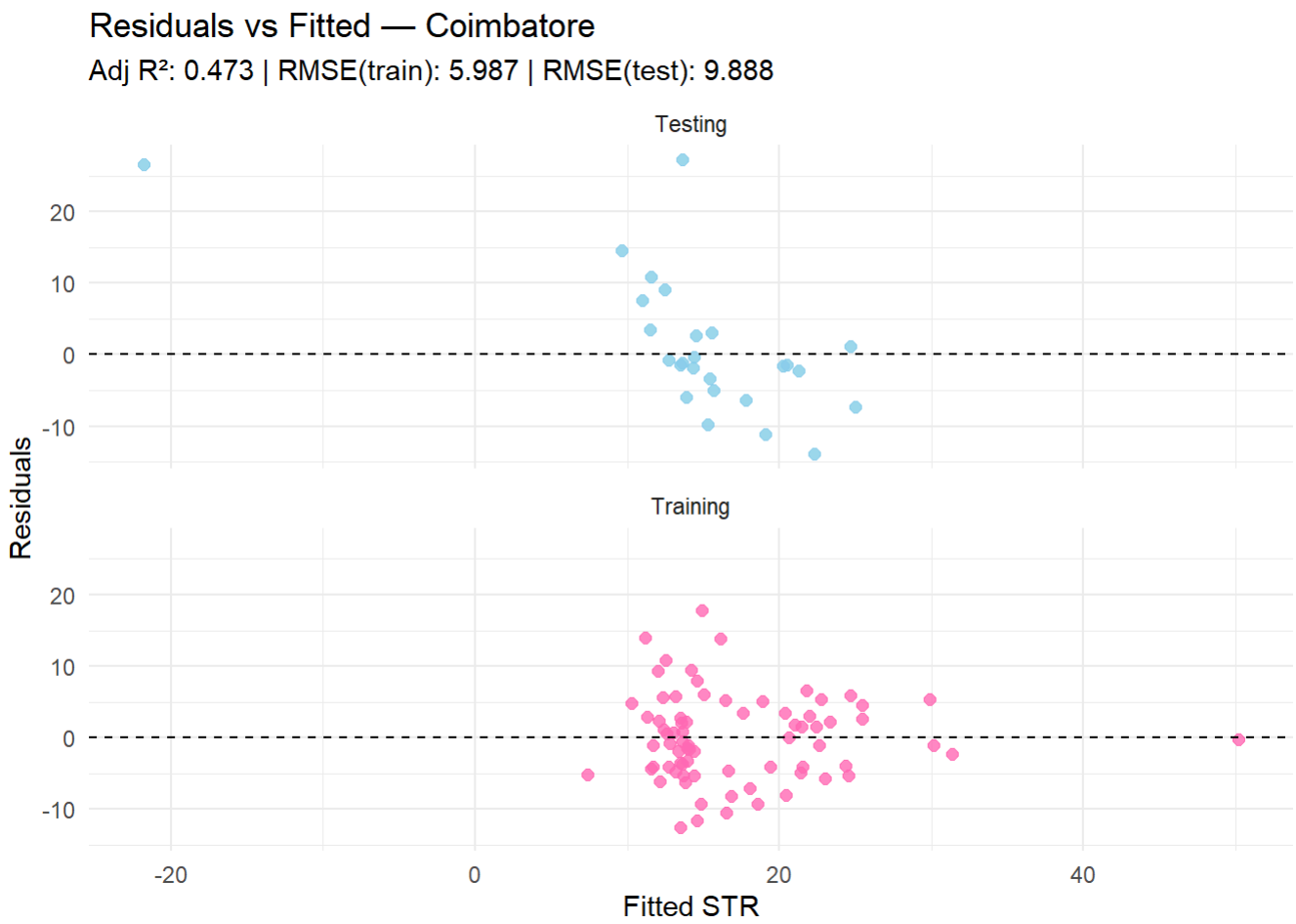

```
## Processing district: dharmapuri
```



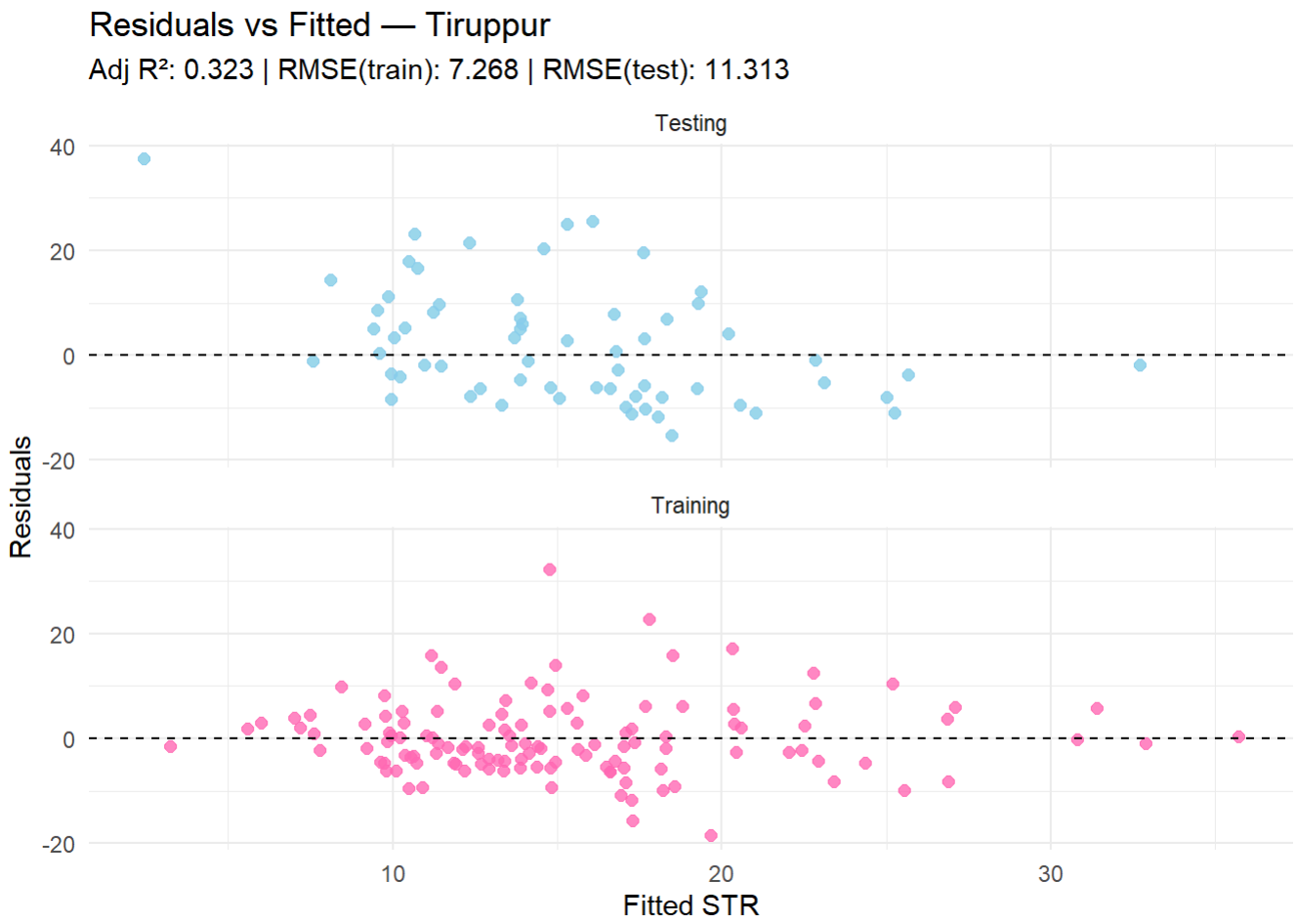

```
## Processing district: krishnagiri
```



```
## Processing district: coimbatore
```




```
## Processing district: tiruppur
```




## District-wise Model Performance for STR — Tamil Nadu

Adj-$R^2$, RMSE (Train/Test), Train/Test Sample Split

| district | n_train | n_test | adj_r2_train | rmse_train | rmse_test |
|---|---|---|---|---|---|
| thiruvallur | 270 | 109 | 0.09789789 | 16.560430 | 19.957389 |
| kancheepuram | 339 | 147 | 0.08087518 | 10.177352 | 10.071068 |
| vellore | 233 | 83 | 0.10044251 | 17.804164 | 14.708886 |
| tiruvannamalai | 427 | 186 | 0.11153665 | 16.699130 | 17.768339 |
| viluppuram | 567 | 247 | 0.13260306 | 11.471944 | 13.727413 |
| salem | 244 | 90 | 0.10056856 | 12.395510 | 9.400932 |
| namakkal | 149 | 53 | 0.21469479 | 6.364178 | 7.204231 |
| erode | 96 | 53 | 0.01665186 | 21.409530 | 12.650225 |
| the nilgiris | 18 | 13 | NaN | 0.000000 | 46.663160 |
| dindigul | 141 | 67 | 0.17004231 | 11.763720 | 24.397656 |
| karur | 80 | 40 | 0.36924874 | 10.277486 | 13.846368 |

## District-wise Model Performance for STR — Tamil Nadu

Adj-$R^2$, RMSE (Train/Test), Train/Test Sample Split

| district | n_train | n_test | adj_r2_train | rmse_train | rmse_test |
|---|---|---|---|---|---|
| tiruchirappalli | 232 | 97 | 0.11523210 | 11.328031 | 13.473392 |
| perambalur | 71 | 31 | 0.22854844 | 8.812478 | 15.588423 |
| ariyalur | 91 | 37 | 0.28909235 | 12.148144 | 15.411610 |
| cuddalore | 356 | 159 | 0.07441226 | NA | 14.053092 |
| nagapattinam | 298 | 126 | 0.04712043 | 17.704746 | 14.775957 |
| thiruvarur | 264 | 127 | 0.07190625 | 18.192711 | 8.734076 |
| thanjavur | 368 | 175 | 0.08408256 | 9.858264 | 16.383677 |
| pudukkottai | 299 | 161 | 0.08714533 | 8.314286 | 11.965713 |
| sivaganga | 129 | 58 | 0.14434395 | 8.660591 | 8.247120 |
| madurai | 206 | 86 | 0.11209067 | 19.095296 | 22.890759 |
| theni | 24 | 13 | 0.36824771 | 4.815053 | 39.761452 |
| virudhunagar | 202 | 73 | 0.08873322 | 18.354331 | 7.941842 |
| ramanathapuram | 157 | 64 | 0.14270126 | 7.885185 | 12.174739 |
| thoothukkudi | 197 | 76 | 0.02487974 | 13.858404 | 26.164876 |
| tirunelveli | 172 | 65 | 0.29458626 | 26.213653 | 16.057855 |
| kanniyakumari | 19 | 6 | NaN | 0.000000 | 225.894703 |
| dharmapuri | 216 | 84 | 0.04681374 | 20.800343 | 12.284515 |
| krishnagiri | 231 | 100 | 0.14332431 | 13.369812 | 14.164580 |
| coimbatore | 75 | 26 | 0.47314651 | 5.987406 | 9.887930 |
| tiruppur | 124 | 64 | 0.32330179 | 7.268287 | 11.313133 |

Stepwise regression, a variable-selection method employed in multiple regression, is the one to automatically determine the predictors that are to be included in the model. Whenever you have a large number of independent variables and wish to arrive at a model that is simpler and more efficient, stepwise regression is helpful. Step-wise first builds a model and then reduces it by taking away or adding predictors one after the other based on their statistical significance—most often, through the use of p-values, AIC, or BIC. It helps in locating less complex and smaller models, getting rid of irrelevant factors and preventing overfitting.

Here, we have used step-wise for different districts with all the variables with respect to the Student Teacher-Ratio. We can see that in most datasets, the test datasets show a similar pattern as the test data-sets. This shows that the model chosen is a good fit with respect to this aspect. Also,among the models obtained the districts that had the best fit models were Thiruvalur, Salem, Cuddalore and Nagapattinam. While the weakest fit models were in the districts, Nammakal, Tiruchirapalli, Ramanathapuram and Coimbatore. The values for this district were very off and the residuals were high. Some models that showed a little arbitrary behaviour were The Nilgiris, Thothukoodi, Kanyakumari and Theni, these were owed to the missing values in this district. We thought this was very confusing as districts like Kanyakumari have a significant population and the missing data leads to a gap in information. This can be misleading and be interpreted incorrectly.

So, even though step-wise here has helped with reducing insignificant variables and arriving at an efficient model, due to such a huge range of variables and because so many variables are unrelated to the Student-Teacher ratio, this can explain the residuals and underfitting. This can be one of the reasons for explaining the models we receive from doing a Step-wise regression for the variables of each district in the state of Tamil Nadu.

#Part B: State-Level Regression with District Effects

```
##
## Call:
## lm(formula = fml_state, data = tn_data_scaled)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.197  -7.276  -1.838   4.383 164.427
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     22.75579    1.19617  19.024  < 2e-16 ***
## total_hhd_availing_pension_under  0.68004    0.17335   3.923 8.82e-05 ***
## sub_centre                         0.61771    0.15046   4.106 4.07e-05 ***
## total_no_of_pregnant_women         0.40545    0.27296   1.485 0.137471
## total_no_of_pregnant_women_recei -0.07399    0.28656  -0.258 0.796263
## phc                                0.48256    0.14747   3.272 0.001071 **
## total_childs_aged_0_to_3_years_r   0.36316    0.20339   1.786 0.074208 .
## availability_of_high_school        1.53744    0.18594   8.269  < 2e-16 ***
## availability_of_middle_school      0.24475    0.15697   1.559 0.118986
## availability_of_mid_day_meal_sch   0.02224    0.14429   0.154 0.877497
## any_primary_sch_toilet             0.27555    0.14445   1.908 0.056472 .
## is_aanganwadi_centre_available     0.29062    0.14466   2.009 0.044563 *
## availability_of_ssc_school        -0.15188    0.18082  -0.840 0.400946
## closed_drainage                    0.12688    0.14401   0.881 0.378335
## is_post_office_available           0.30963    0.16356   1.893 0.058374 .
## total_hhd_having_piped_water_con   0.36624    0.17421   2.102 0.035559 *
## regular_market                    -0.06751    0.14489  -0.466 0.641278
## total_trained_trainee_under_skil   0.33578    0.14897   2.254 0.024218 *
## no_of_children_not_attending_sch   0.60888    0.14394   4.230 2.36e-05 ***
## total_no_of_elected_representati   0.26412    0.14269   1.851 0.064190 .
## total_hhd_with_clean_energy       -0.10263    0.19122  -0.537 0.591471
## availability_of_railway_station   -0.27962    0.14582  -1.917 0.055206 .
## total_childs_categorized_non_stu   0.07013    0.14885   0.471 0.637564
## district_namecoimbatore           -7.27524    1.80081  -4.040 5.39e-05 ***
## district_namecuddalore             0.83898    1.33834   0.627 0.530752
## district_namedharmapuri           -3.63604    1.42939  -2.544 0.010983 *
## district_namedindigul             -4.32077    1.52194  -2.839 0.004536 **
## district_nameerode                -2.42421    1.63123  -1.486 0.137283
## district_namekancheepuram         -2.61889    1.35022  -1.940 0.052460 .
## district_namekanniyakumari        -3.32130    2.97891  -1.115 0.264907
## district_namekarur                -4.05746    1.71536  -2.365 0.018033 *
## district_namekrishnagiri          -3.35323    1.40505  -2.387 0.017028 *
## district_namemadurai              -1.45118    1.43367  -1.012 0.311462
## district_namenagapattinam         -3.12241    1.36037  -2.295 0.021741 *
## district_namenamakkal             -8.82991    1.52754  -5.780 7.70e-09 ***
## district_nameperambalur           -3.00317    1.79050  -1.677 0.093523 .
## district_namepudukkottai          -2.23346    1.35584  -1.647 0.099534 .
## district_nameramanathapuram       -6.95434    1.50030  -4.635 3.61e-06 ***
## district_namesalem                -3.38570    1.40337  -2.413 0.015862 *
## district_namesivaganga            -6.97642    1.55213  -4.495 7.05e-06 ***
## district_namethanjavur            -3.59039    1.33460  -2.690 0.007153 **
## district_namethe nilgiris        -12.96865    2.73715  -4.738 2.19e-06 ***
## district_nametheni                -8.20054    2.51727  -3.258 0.001127 **
## district_namethiruvallur          -1.44240    1.38508  -1.041 0.297726
## district_namethiruvarur           -2.91633    1.37744  -2.117 0.034269 *
```

```
## district_namethoothukkudi         -5.20179    1.44901  -3.590 0.000333 ***
## district_nametiruchirappalli      -3.07880    1.40648  -2.189 0.028622 *
## district_nametirunelveli          -2.20008    1.48364  -1.483 0.138137
## district_nametiruppur             -7.91723    1.56077  -5.073 4.00e-07 ***
## district_nametiruvannamalai        1.60218    1.31755   1.216 0.224006
## district_namevellore              -2.38870    1.41629  -1.687 0.091716 .
## district_nameviluppuram           -1.12437    1.28997  -0.872 0.383438
## district_namevirudhunagar         -3.99224    1.45485  -2.744 0.006080 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.42 on 8947 degrees of freedom
##   (3 observations deleted due to missingness)
## Multiple R-squared:  0.06896,    Adjusted R-squared:  0.06355
## F-statistic: 12.74 on 52 and 8947 DF,  p-value: < 2.2e-16
```

Part B — State-Level Regression Coefficients (Table 1)

| Variable | Estimate | Std_Error | statistic | P_Value |
|---|---|---|---|---|
| (Intercept) | 22.7558 | 1.1962 | 19.023938 | 0.0000 |
| the nilgiris | −12.9687 | 2.7372 | -4.738008 | 0.0000 |
| namakkal | −8.8299 | 1.5275 | -5.780462 | 0.0000 |
| theni | −8.2005 | 2.5173 | -3.257718 | 0.0011 |
| tiruppur | −7.9172 | 1.5608 | -5.072659 | 0.0000 |
| coimbatore | −7.2752 | 1.8008 | -4.039976 | 0.0001 |
| sivaganga | −6.9764 | 1.5521 | -4.494756 | 0.0000 |
| ramanathapuram | −6.9543 | 1.5003 | -4.635290 | 0.0000 |
| thoothukkudi | −5.2018 | 1.4490 | -3.589897 | 0.0003 |
| dindigul | −4.3208 | 1.5219 | -2.838988 | 0.0045 |
| karur | −4.0575 | 1.7154 | -2.365370 | 0.0180 |
| virudhunagar | −3.9922 | 1.4549 | -2.744086 | 0.0061 |
| dharmapuri | −3.6360 | 1.4294 | -2.543765 | 0.0110 |
| thanjavur | −3.5904 | 1.3346 | -2.690242 | 0.0072 |
| salem | −3.3857 | 1.4034 | -2.412546 | 0.0159 |
| krishnagiri | −3.3532 | 1.4050 | -2.386557 | 0.0170 |
| kanniyakumari | −3.3213 | 2.9789 | -1.114938 | 0.2649 |

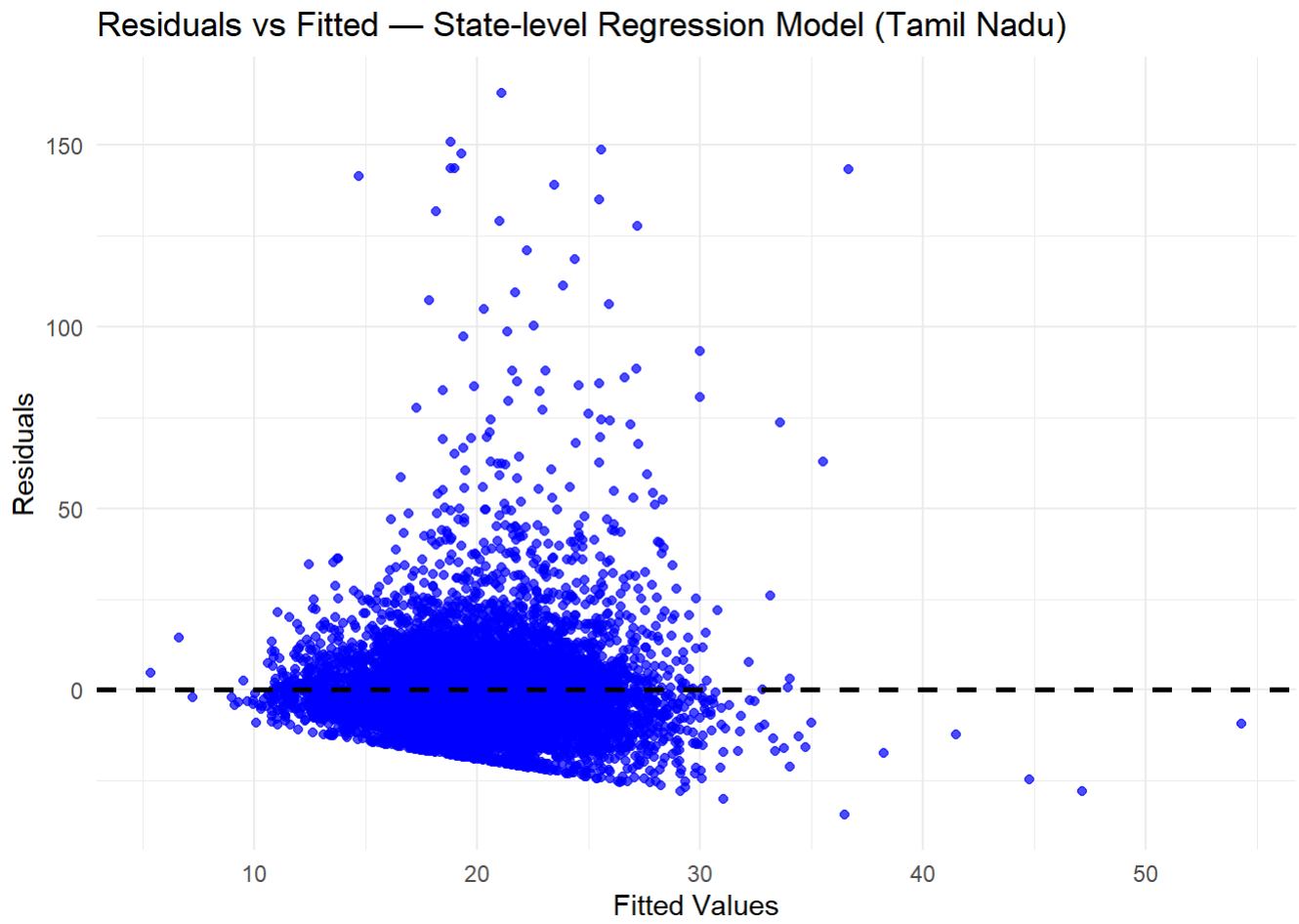
| Part B — State-Level Regression Coefficients (Table 1) | | | | |
|---|---|---|---|---|
| Variable | Estimate | Std_Error | statistic | P_Value |
| nagapattinam | −3.1224 | 1.3604 | -2.295264 | 0.0217 |
| tiruchirappalli | −3.0788 | 1.4065 | -2.189014 | 0.0286 |
| perambalur | −3.0032 | 1.7905 | -1.677278 | 0.0935 |
| thiruvarur | −2.9163 | 1.3774 | -2.117214 | 0.0343 |
| kancheepuram | −2.6189 | 1.3502 | -1.939597 | 0.0525 |
| erode | −2.4242 | 1.6312 | -1.486119 | 0.1373 |
| vellore | −2.3887 | 1.4163 | -1.686597 | 0.0917 |
| pudukkottai | −2.2335 | 1.3558 | -1.647287 | 0.0995 |
| tirunelveli | −2.2001 | 1.4836 | -1.482896 | 0.1381 |
| tiruvannamalai | 1.6022 | 1.3176 | 1.216028 | 0.2240 |

| Part B — State-Level Regression Coefficients (Table 2) | | | | |
|---|---|---|---|---|
| Variable | Estimate | Std_Error | statistic | P_Value |
| availability_of_high_school | 1.5374 | 0.1859 | 8.2685190 | 0.0000 |
| madurai | −1.4512 | 1.4337 | -1.0122156 | 0.3115 |
| thiruvallur | −1.4424 | 1.3851 | -1.0413828 | 0.2977 |
| viluppuram | −1.1244 | 1.2900 | -0.8716211 | 0.3834 |
| cuddalore | 0.8390 | 1.3383 | 0.6268829 | 0.5308 |
| total_hhd_availing_pension_under | 0.6800 | 0.1734 | 3.9228329 | 0.0001 |
| sub_centre | 0.6177 | 0.1505 | 4.1055531 | 0.0000 |
| no_of_children_not_attending_sch | 0.6089 | 0.1439 | 4.2301900 | 0.0000 |
| phc | 0.4826 | 0.1475 | 3.2723588 | 0.0011 |
| total_no_of_pregnant_women | 0.4055 | 0.2730 | 1.4854066 | 0.1375 |
| total_hhd_having_piped_water_con | 0.3662 | 0.1742 | 2.1022546 | 0.0356 |
| total_childs_aged_0_to_3_years_r | 0.3632 | 0.2034 | 1.7855363 | 0.0742 |
| total_trained_trainee_under_skil | 0.3358 | 0.1490 | 2.2540360 | 0.0242 |

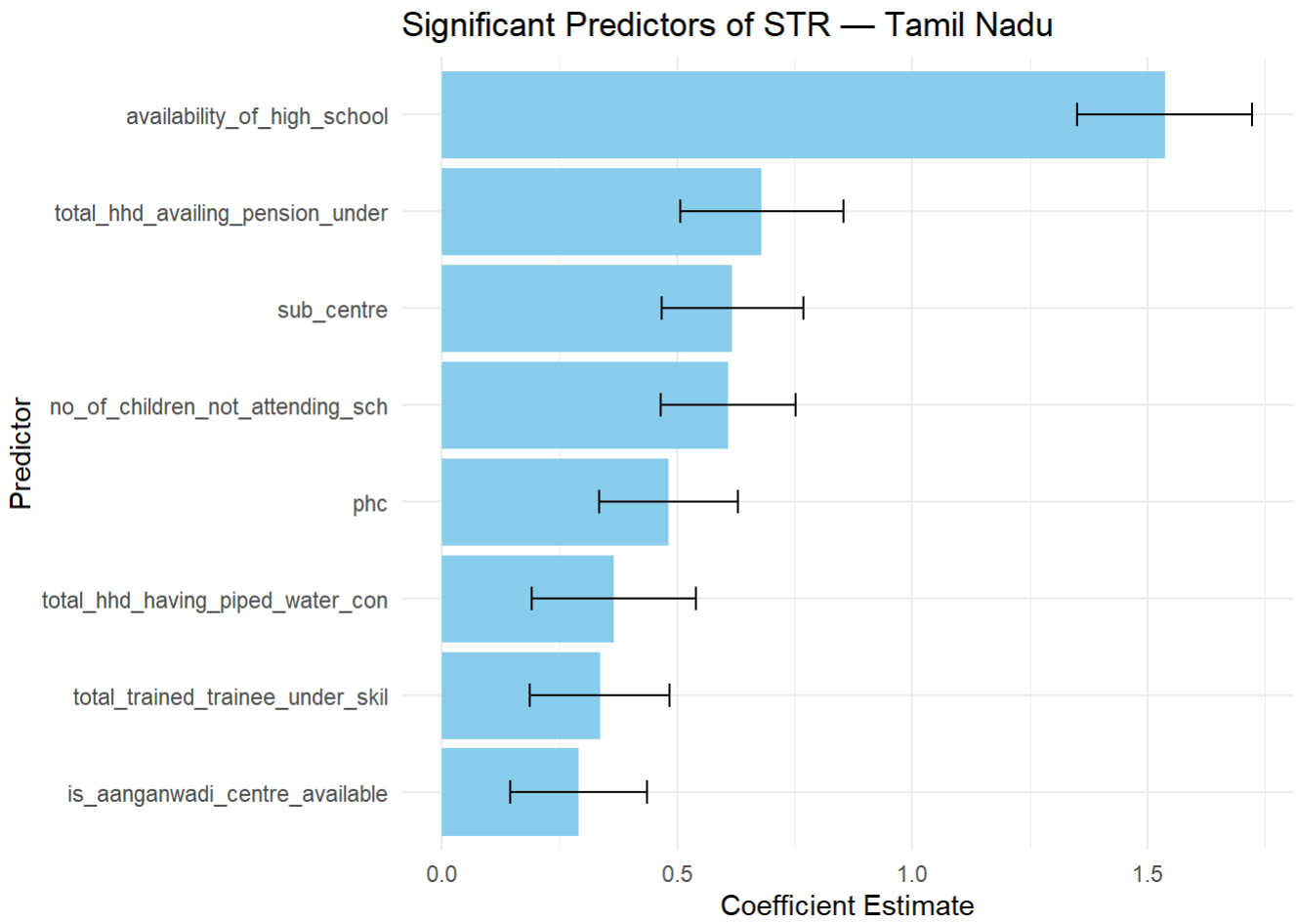
| Part B — State-Level Regression Coefficients (Table 2) | | | | |
|---|---|---|---|---|
| Variable | Estimate | Std_Error | statistic | P_Value |
| is_post_office_available | 0.3096 | 0.1636 | 1.8931213 | 0.0584 |
| is_aanganwadi_centre_available | 0.2906 | 0.1447 | 2.0090398 | 0.0446 |
| availability_of_railway_station | −0.2796 | 0.1458 | -1.9174983 | 0.0552 |
| any_primary_sch_toilet | 0.2756 | 0.1444 | 1.9076200 | 0.0565 |
| total_no_of_elected_representati | 0.2641 | 0.1427 | 1.8510868 | 0.0642 |
| availability_of_middle_school | 0.2448 | 0.1570 | 1.5591936 | 0.1190 |
| availability_of_ssc_school | −0.1519 | 0.1808 | -0.8399737 | 0.4009 |
| closed_drainage | 0.1269 | 0.1440 | 0.8810123 | 0.3783 |
| total_hhd_with_clean_energy | −0.1026 | 0.1912 | -0.5367247 | 0.5915 |
| total_no_of_pregnant_women_recei | −0.0740 | 0.2866 | -0.2581944 | 0.7963 |
| total_childs_categorized_non_stu | 0.0701 | 0.1488 | 0.4711232 | 0.6376 |
| regular_market | −0.0675 | 0.1449 | -0.4659285 | 0.6413 |
| availability_of_mid_day_meal_sch | 0.0222 | 0.1443 | 0.1541474 | 0.8775 |

District dummies were created to check how the average student-teacher ratio in each district differs from the chosen reference districts after controlling for all other factors at the village level. The omitted district by model here is Ariyalur, so the coefficient of each district is compared with this district. The districts with values close to zero show that their performance doesn't differ much from baseline districts. Many districts, such as The Nilgirs, Tiruppur, Coimbatore and Theni, show a strong negative coefficient, meaning they are consistently having a lower student-teacher ratio compared to Ariyalur, indicating better teacher availability. There are some districts with moderate negative coefficients, such as Dindigul, Karur, Dharmapuri, Thanjavur and Salem. A moderate negative coefficient indicates they have relatively lower STR than Ariyalur. The only district that showed a very high positive coefficient is Tiruvannamalai, but it is statistically insignificant. Overall, the district dummy results confirm that geographical location has an independent influence on STR, as differences persist across districts even after accounting for village-level predictors.

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




The residual vs. fitted plot is not well distributed around zero, indicating that a significant portion of the variation remains unexplained by the model. It predicts reasonably well for villages with lower STR, but as STR increases, the residuals become more scattered. This indicates that the model became less reliable for villages with higher STR. In some villages, STR is higher than expected, indicating that the model under-predicts STR. On the other hand, for some villages, over-prediction occurs where fitted values tend to be smaller. Some villages are outliers, which don't follow the pattern at all. This is because there could be special circumstances like seasonal migration of students, which temporarily increases the number of students, budget constraints and poor record keeping. The model does not recognise these special situations since the chosen variables do not include them, which is why some changes in the student–teacher ratio remain unexplained.

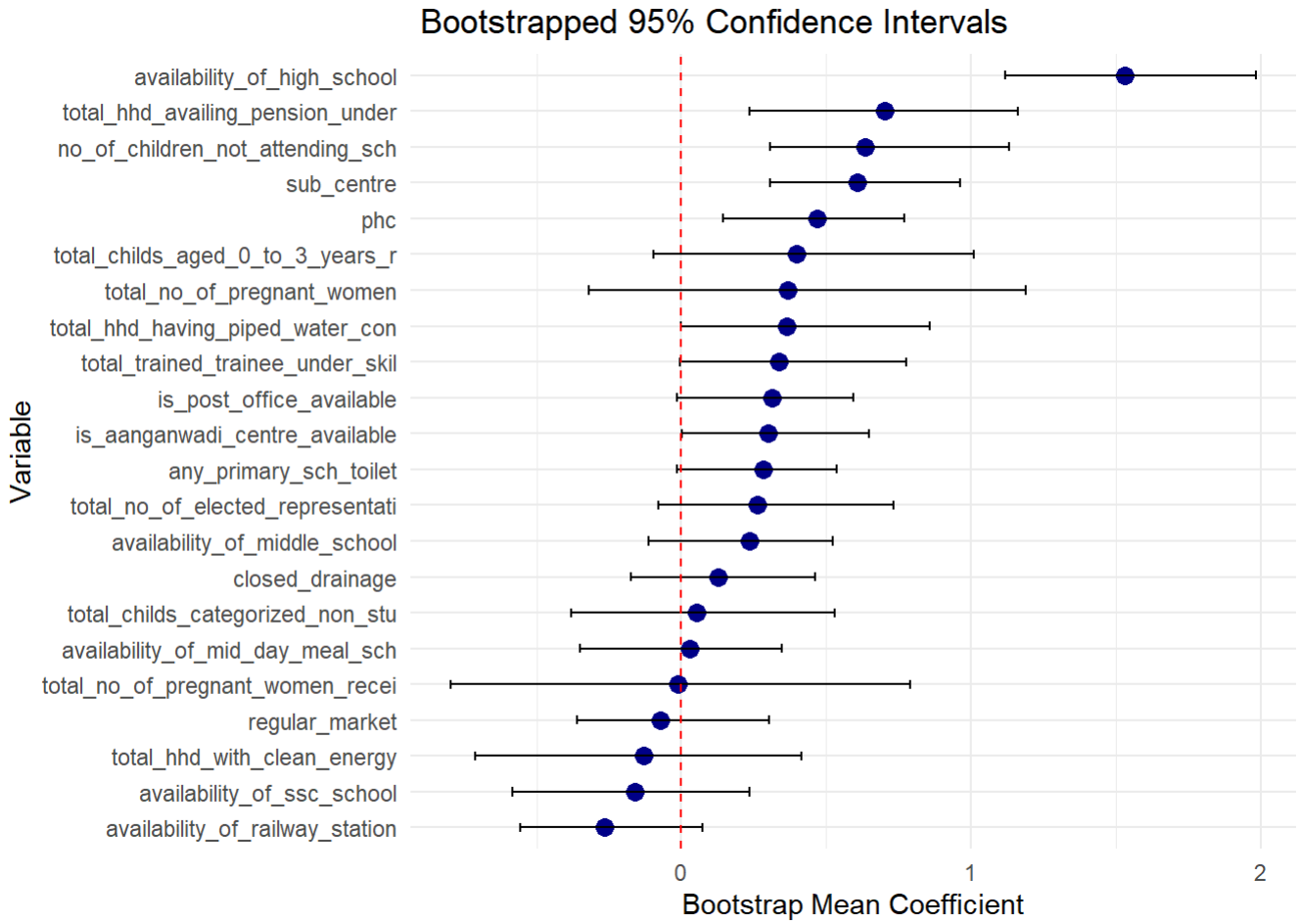



| term | estimate | std.error | statistic | p.value |
|---|---|---|---|---|
| total_hhd_availing_pension_under | 0.6800373 | 0.1733536 | 3.922833 | 0.0000882 |
| sub_centre | 0.6177147 | 0.1504583 | 4.105553 | 0.0000407 |
| phc | 0.4825647 | 0.1474669 | 3.272359 | 0.0010706 |
| availability_of_high_school | 1.5374436 | 0.1859394 | 8.268519 | 0.0000000 |
| is_aanganwadi_centre_available | 0.2906222 | 0.1446573 | 2.009040 | 0.0445629 |
| total_hhd_having_piped_water_con | 0.3662366 | 0.1742114 | 2.102255 | 0.0355589 |
| total_trained_trainee_under_skil | 0.3357801 | 0.1489684 | 2.254036 | 0.0242180 |
| no_of_children_not_attending_sch | 0.6088801 | 0.1439368 | 4.230190 | 0.0000236 |

The graph shows that the significant state-level predictor influences the STR after controlling for district-level differences. The availability of high schools has a strong positive influence on the student-teacher ratio, meaning the areas with high schools tend to have higher STR. This is because villages with high schools often act as centres for nearby villages, so not only their own population but also people from nearby villages come to those schools. This also increases the number of primary students, but teachers don't increase at the same rate, resulting higher STR. With that, other variables also show a positive association with are presence of subcenters and PHCs, total households availing pension and the number of children not attending school. As highlighted in studies such as the Karnataka primary health and elementary education review, government service uptake is strongly influenced by where facilities are located, the visibility of amenities, and the clustering of essential services in urban and peri-urban areas. PHCs and subcenters are often established in

villages that have higher population densities and stronger administrative systems. These conditions attract more families, and when more families are settled student enrollment rate is higher, which increases the student-teacher ratio.

```
##
##
## Table:  Bootstrapped 95% CI for STR Predictors (Tamil Nadu)
##
## |Variable                         |       Mean|   Lower_CI|  Upper_CI|
## |:--------------------------------|----------:|----------:|---------:|
## |any_primary_sch_toilet           |  0.2822909| -0.0159665| 0.5361280|
## |availability_of_high_school      |  1.5292708|  1.1181708| 1.9817121|
## |availability_of_mid_day_meal_sch |  0.0292379| -0.3486208| 0.3464825|
## |availability_of_middle_school    |  0.2356176| -0.1122659| 0.5210316|
## |availability_of_railway_station  | -0.2643939| -0.5554603| 0.0718674|
## |availability_of_ssc_school       | -0.1603160| -0.5816675| 0.2352821|
## |closed_drainage                  |  0.1272100| -0.1732359| 0.4627407|
## |is_aanganwadi_centre_available   |  0.2996648|  0.0019620| 0.6485552|
## |is_post_office_available         |  0.3138315| -0.0165405| 0.5922777|
## |no_of_children_not_attending_sch |  0.6350192|  0.3044202| 1.1291289|
## |phc                              |  0.4670381|  0.1422583| 0.7688241|
## |regular_market                   | -0.0715448| -0.3594138| 0.3017281|
## |sub_centre                       |  0.6051618|  0.3066123| 0.9626415|
## |total_childs_aged_0_to_3_years_r |  0.3986430| -0.0959265| 1.0098738|
## |total_childs_categorized_non_stu |  0.0528360| -0.3786045| 0.5294690|
## |total_hhd_availing_pension_under |  0.7012036|  0.2342967| 1.1594900|
## |total_hhd_having_piped_water_con |  0.3628353| -0.0015928| 0.8563226|
## |total_hhd_with_clean_energy      | -0.1283285| -0.7104944| 0.4125162|
## |total_no_of_elected_representati |  0.2606714| -0.0792524| 0.7308906|
## |total_no_of_pregnant_women       |  0.3673609| -0.3206306| 1.1877795|
## |total_no_of_pregnant_women_recei | -0.0123650| -0.7969580| 0.7899115|
## |total_trained_trainee_under_skil |  0.3353143| -0.0047938| 0.7765240|
```




The bootstrapping CI graph plot shows which variable matters for explaining the STR, because the confidence bars do not cross the zero line in the middle. This graph can be a good indicator for the people in power to know which segment needs more investment. This graph shows which predictors are consistently significant even after sample variations. Bootstrapping offers a more robust sense of uncertainty by continually resampling the data. The predictors whose confidence intervals do not cross zero can be considered statistically reliable. The variables whose confidence interval crosses zero indicate that their impact on the student-teacher ratio is not consistent; it changes depending on the sample. This also tells us these are not strong drivers or stable drivers of STR at the state level. The variable that stands out very clearly and has the most stable positive effect is the availability of high school. Again, indicating villages with high schools tend to have higher student-teacher ratios as discussed earlier. In a similar vein, places with greater rates of pension-receiving households, out-of-school children, and basic health facilities have higher STR, which is indicative of underlying pressure factors such as service-seeking load, household vulnerabilities, and population demand. There are a few variables such as regular_market, total_hhd_with_clean_energy, and availability_of_railway_station, whose CIs cross zero and somewhat fall below zero. This indicates that villages with access to formal markets, better connectivity and cleaner household energy tend to have relatively lower STR. The bootstrapped CIs graph shows the significance of the OLS predictors and thus also means that the coefficients are comparatively stable, even after the observed heteroscedasticity

The results from our findings could be misleading, as there are other factors one must look into while understanding student teacher ratios. An Indian Express news article dated 29th October 2025 wrote about a sharp increase in dropout rates in Tamil Nadu's schools. According to UDISE+, in primary, the drop out rate rose from 0% to 2.7% in 2024-25. This in turn could be a reason for a lower STR in some district. Nilgiris, being a hilly area, has a scattered population and thus it is more likely the region has more number of single teacher schools. Since the area has a lot of migrants and labourers, it is also likely that the dropout rates are higher as well. Considering these possible factors, we cannot conclude that the low STRs in Nilgiris is a good thing. Based on UDISE reports, The Pupil Teacher Ratio is at 24 now, but the number of single teacher schools have increased sharply- 2758 in 2023 to 3671. The high STR in these schools is a gross violation of Section 25 of the Right of Children for Free and Compulsory Education Act (RTE). As per the act, the ratio should be 30:1 per

school at the primary level. A 2018 Times of India report showed Krishnagiri and Vellore as backward in terms of STR as a total of 3800 teacher posts were left vacant in government and government-aided schools combined. A possible reason for the high STR in areas like Villupuram could be due to its population size, being so close to Pondicherry, Villupuram houses a lot of migrants as well. Here, even though Student Enrollment is high, the issue of less number of teachers.

Conslusion: Overall, we can conclude that although several variables showed statistically significant associations with the student–teacher ratio, the explanatory power of the model remains low. This indicates that STR in Tamil Nadu is shaped by many additional structural, administrative and social factors that are not captured in the current dataset. It is also important to note that the study does not establish causation; therefore, none of the variables can be said to directly lead to a high or low student–teacher ratio. The pattern of significant predictors suggests that in high-population and service-dense areas, educational demand may be rising faster than teacher deployment, creating pressure on existing school resources. Future research should incorporate additional dimensions — such as dropout rates, seasonal migration and socioeconomic mobility — to obtain a more holistic understanding of the drivers of STR across the state. A deeper investigation using richer datasets and more advanced modelling techniques will be required to fully understand the real patterns underlying teacher allocation and student distribution.

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